

REACH: A Reproducible Benchmark for Rare Malignant Cell Detection in Single Cell RNA Sequencing

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1 REACH: A Reproducible Benchmark for Rare Malignant-Cell Detection in Single-Cell RNA Sequencing

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1.1 Abstract

Rare malignant cells in ~~single-cell~~ single-cell RNA sequencing (scRNA-seq) ~~span~~ can represent circulating tumor cells, minimal residual disease, ~~drug-tolerant persisters, and small malignant~~ drug-tolerant persisters, metastatic intermediates, or small tumor compartments embedded in ~~heterogeneous tumor microenvironments~~. ~~To detect rare cells, existing methods have been evaluated inconsistently because studies differ in datasets, label definitions, prevalence regimes, score contracts, and failure handling. Here, we present a complex microenvironment.~~ REACH targets the specific problem of **rare malignant-cell detection in a heterogeneous tumour microenvironment (TME)**: positives are malignant cells embedded in an immune/stromal background, and the task is **imbalanced cell-level ranking**, not intra-lineage sub-state discovery. REACH therefore evaluates how well methods surface scarce malignant cells above an abundant TME background under controlled prevalence, synthetic stress, null-control, and label-noise conditions. These populations are biologically important but statistically difficult: the positive class is scarce, labels are assembled from imperfect evidence, and a high AUROC can coexist with poor recovery of top-ranked cells. We present REACH, not a new detector but a reproducible benchmark ~~framework, Rare-cell Evaluation Across Cancer Heterogeneity (REACH), that treats rare malignant-cell detection as an imbalanced ranking task. REACH combines confidence-tiered multi-arm labels, five complementary and software/data resource for rare malignant-cell detection that treats the task as imbalanced cell-level ranking.~~ REACH v1.2.0 combines ten curated public scRNA-seq datasets (approx. 360,000 cells), six evaluation tracks (~~controlled real spike-ins, synthetic splatter stress, null background controls, natural blood/CTC prevalence, and supervised label noise diagnostics~~), A–F), ten standardized method wrappers, ~~fallback-aware~~ confidence-tiered multi-arm labels, fallback-aware metrics, nonparametric paired statistics, and figure regeneration from ~~audited-frozen~~ result tables. ~~In its current form, REACH evaluates ten recent computational methods on ten curated scRNA-seq cancer datasets comprising 359,739 cells. Across 1,110 benchmark tasks, this produces 125 benchmark units, REACH produced 11,100 individual method performance measurements. The supervised highly variable gene logistic regression ceiling achieves near perfect separation for many tumor related labels, with a 250 method-unit evaluations. On the primary controlled real-cell Track A endpoint, the supervised in-sample hvg_logreg ceiling reached median average precision of 1.0, indicating clear gene expression differences between tumor related and non tumor cell populations. In contrast, current unsupervised ranking methods recover only part of these differences (AP) 1.000 (mean 0.909, mean AUROC 0.953), showing that high-confidence labels often retain recoverable expression signal. The exploratory comparator CaSee obtains a median average precision of CaSee reached median AP 0.512, whereas FiRE, the strongest ranked published detector, achieves a median average precision of but is reported separately from ranked published detectors. Among ranked published methods, FiRE had the highest primary median AP (0.313. Statistical testing confirms substantial performance differences among methods on the common benchmark subset. More importantly, REACH shows that performance is highly context dependent: no single~~

method consistently performs best across tumor types, and method behavior under ranking, null controls, and label noise cannot be adequately represented by a single summary score. These findings highlight a clear-), while paired tests showed it was not significantly different from `expr_threshold` or `scMalignantFinder`. Global Track A method differences were strongly supported on the 140-unit common subset (Friedman $\chi^2=695.05$, $p=8.0\times 10^{-1}$; Iman-Davenport $F=171.01$, $p=8.4\times 10^{-211}$). REACH exposes a large gap between supervised expression-based separability and the current performance of unsupervised rare malignant cell detection methods. All code, data, figures, and documentation required to reproduce the analysis are available in the public GitHub repository:-

2 Introduction

Tumor samples are often highly mixed. Along with dominant malignant and non-malignant populations, they may contain small groups of cells that are biologically important but difficult to observe in bulk measurements. Single-cell RNA sequencing (scRNA-seq) allows us to look at such populations of cells separately, and to investigate tumor heterogeneity with an unprecedented level of detail [?, ?]. This is especially important when it comes to circulating tumor cells, metastatic intermediates, minimal residual disease cells, drug-tolerant persisters and small malignant compartments in the tumor microenvironment. These populations can be low-frequency, *e.g.*, contribute to 1% or less in a given dataset [?, ?]. That not only makes them challenging to identify in biological terms, but also in statistical ones. They can be more difficult to extract if rare cells are in a more diluted cluster. If these cells are rare, mixed with neighbouring cells expressing related cell types or have low confidence from expression and copy number evidence, they become nearly impossible to identify [?, ?, ?] separability and current unsupervised ranking, substantial prevalence sensitivity, dataset-specific winners, and method reliability issues that would be hidden by a single leaderboard number.

This is particularly challenging for cancer data sets where the label malignant may not be clear-cut. A cell can be classified as malignant based on copy number alteration, cancer associated expression signatures, somatic mutation evidence, neighborhood structure or expert annotations [?, ?, ?]. However, in practice, the evidential sources may not be fully congruent. For some cells, there is strong evidence of expression but weak support by copy number, while others can be hard to classify if they are near non malignant populations. This sets rare malignant cell detection apart from its cousin cell type annotation, where the expected classes are usually more clearly delineated. Moreover, the problem is extremely imbalanced as there are frequently many more background cells surrounding the malignant cells of interest [?, ?].

This imbalance also raises questions about how methods should be assessed. The Area Under the Receiver Operating Characteristic Curve (AUROC) is widely used as a performance metric for machine learning classification and ranking tasks, but it becomes unwieldy and confusing if the positive class size shrinks very small. If true rare malignant cells are not ranked anywhere near the top of the list for a method, it may obtain a quite reasonable AUROC. Due to the high imbalance ratio, precision recall based measures are often more meaningful in this setting since they reflect whether rare positive cells are recovered among the highest ranked predictions [?, ?]. However, performance should never be evaluated on just a single statistic. Changing label definition, data distribution, null behavior, method failure and statistical error makes how to interpret benchmark results a complex matter.-

There have been multiple scRNA-seq benchmarks that have already been created to test batch effect correction, clustering, cell type annotation and rare cell discovery. These benchmarks serve as useful reference standards for method comparison. Nevertheless, rare malignant cell detection requires a more tailored assessment plan. The targeted population may be very small, the label may require multiple types of biological evidence, and the same approach may not perform uniformly across tumor types, sequencing platforms or prevalence settings [?, ?, ?, ?]. - Thus, a benchmark for this task should not return just one best method. It should use common rules to compare methods, separate supervised reference performance from unsupervised ranking performance, log failed or unstable runs, include null control analysis, and report results in a cancer and dataset specific manner[?].-

We herein present *Rare cell Evaluation Across Cancer Heterogeneity* (REACH), a reproducible benchmark for rare malignant cell detection in scRNA-seq data (Fig. ??). REACH formulates the detection of rare malignant cells as an imbalanced ranking problem. It is to expect that a method would give more scores to high-confidence malignant cells than to high confidence background cells. It contains the various stages of a typical pipeline, from data ingestion and preprocessing to multi-evidence label construction, evaluation track generation, method execution, metric summarization and statistical analysis.-

1.1 Public Release Scope

Five evaluation tracks were designed to analyze different aspects of method behavior in REACH (Fig. ??). Track A evaluates real malignant cells under controlled rarity levels and serves as the primary ranking endpoint. Track B tests method performance under synthetic stress. Track C uses background-dominated controls to assess false-positive behavior when rare malignant cells are not expected. Track D evaluates natural prevalence in blood and circulating tumor cell datasets. Track E examines how the supervised ceiling changes when labels are deliberately corrupted; therefore, it is interpreted as a supervised label noise diagnostic rather than as a general robustness test for unsupervised methods. The current audited REACH snapshot includes 10 datasets, 10 methods, and 100 individual method performance measurements. The framework also provides instructions for adding new datasets and methods, so that it can be extended in future studies.

The results show that many high confidence malignant and background labels contain gene-expression differences that can be recovered by a supervised model. However, current unsupervised and rarity-based methods recover only part of this separation on the primary Track A endpoint [?, ?]. REACH also shows that method performance changes across datasets and evaluation tracks. A method that performs well in one tumor context may not remain the strongest method in another [?, ?]. These findings support the need for a reproducible benchmark that evaluates rare malignant cell detection in relation to dataset context, rarity level, failure behavior, and evaluation condition. CSV snapshots, Phase 12 PNG previews, and submission-ready figures. Large processed datasets, track units, full prediction outputs, and complete result archives are external Zenodo/GitHub release assets.

2 Related work

1.1 Rare cell discovery in scRNA-seq

1.1 Contributions

Finder of Rare Entities (FiRE) assigns a density-derived rarity score to each cell and is included in REACH as one of the ranked published comparators [?]. The Cell Subtype Identification from Up-regulated gene Sets (CellSIUS) method uses up-regulated gene sets and greater specificity for rare sub-populations of complex single cell populations from single-cell RNA sequencing to detect rare populations [?]. The detection of rare populations using single-molecule RNA fluorescence *in-situ* hybridization (RNA-FISH) shows the dual dependence of the detection of rare expression on both the studied gene set and the number of cells from which expression is derived [?]. Further analysis of cellular expression using deep clustering and anomaly detection scorers, such as Deep Single Cell Ensemble Network Analysis (DeepScena) and single-cell Clustering-based Anomaly Detection (scCAD) are starting to give the first steps of multi-sample benchmarking to get an insight into the effect of pool and/or batch correction on the recovery of rare populations [?, ?]. Together, these methods motivate their inclusion in REACH, but they also show why a cancer-specific benchmark is needed: rare malignant cells may not share a single gene program and often require confirmation.

1. REACH formalises **rare malignant-cell detection in heterogeneous tumour microenvironments** as a per-unit, imbalanced ranking task over high-confidence positive malignant cells and background TME cells — explicitly an inter-lineage contrast, not intra-lineage sub-state discovery.
2. It defines confidence-tiered labels from multiple evidence arms [?, ?].

1.2 Single-cell benchmarking

Various mixture control tests have been used to benchmark a number of different scRNA-seq pipelines [?]. Other studies have provided benchmarking of cell type classification, batch corrections, clustering for tumor types in cancer, automated annotation of tumor cells and copy number inference [?, ?, ?, ?, ?]. These studies established useful practices for benchmarking fixed datasets with clearly defined metrics. However, the low frequency of detection for the presence of malignant cells required prevalence-adjusted methods, null controls, fallbacks and ranked metrics, based on the previous template [?, ?, ?].

1.2 Malignant cell annotation and CNV evidence

Labeling of malignant cells is typically done using the combined results of expression and copy number variation (CNV) inference [?]. Copy number Karyotyping of Tumors (CopyKAT) can delineate copy number and clonal substructures [?]. Moreover, single-cell Annotation of Tumor and Immune Cells (scATOMIC) assigns cell types to the cells in the tumor microenvironment for all types of cancer [?]. Here, variant calling

has been tested as another way of providing evidence of the presence of a cancer cell [?]. REACH uses CNV and cancer signature evidence to support label construction rather than to create a separate leaderboard score. In other words, these signals help define high confidence malignant and background cells, but they are not themselves ranked as competing detection methods. Candidate methods that failed to provide valid cell level scores or did not satisfy the required score contract are documented in the method inclusion audit and excluded from the main ranking analyses (Fig. ??(a)).

2 Methodology

1.1 Task definition and metrics

For benchmark unit u , let $X_u \in \mathbb{R}^{n_u \times p}$ be the log normalized expression matrix, where n_u denotes the number of cells and p the number of genes in unit u . Let $y_{ui} \in \{0, 1\}$ denote the ground truth label of cell i in unit u , where $y_{ui} = 1$ indicates a high confidence malignant (rare cell) state and $y_{ui} = 0$ indicates background or non-malignant status. A method m returns a continuous cell level score

$$f_m(X_u) = s_u^{(m)} \in \mathbb{R}^{n_u},$$

where $s_u^{(m)}$ is the vector of predicted scores for all cells in unit u , and larger scores indicate stronger predicted malignant or rare cell status. REACH evaluates ranking performance rather than thresholded diagnosis. For a ranked list, the standard interpolated Average Precision (AP) [?, ?] is

$$\text{AP}(m, u) = \frac{\sum_{k=1}^{n_u} (R_k - R_{k-1}) P_k}{n_u},$$

where P_k and R_k denote precision and recall after the first k ranked cells, respectively, with $R_0 = 0$. The primary method summary is the median Track A AP computed across all valid benchmark units,

$$S_m = \frac{\text{median}_{u \in \mathcal{U}_{A,m}^{\text{valid}}} \text{AP}(m, u)}{n_u},$$

where $\mathcal{U}_{A,m}^{\text{valid}}$ denotes the set of Track A benchmark units that remain after fallback-based filtering for method m . Because raw AP depends on class prevalence, REACH additionally reports prevalence-adjusted quantities. Let

$$\pi_u = \frac{1}{n_u} \sum_i y_{ui}$$

denotes the positive class prevalence in the benchmark unit u . AP above chance is defined as $\text{AP}(m, u) - \pi_u$, and the normalized AP is

$$\text{nAP}(m, u) = \frac{\text{AP}(m, u) - \pi_u}{1 - \pi_u}.$$

AUROC is reported as a secondary discrimination metric [?, ?]. Calibration metrics, including the Brier score and Expected Calibration Error (ECE), are reported only for probability-like outputs and are explicitly marked as *not applicable* for rank-based or anomaly-based scores. F1, precision, recall, balanced accuracy, and Matthews correlation coefficient (MCC) are reported at top k , where k equals the number of positive cells in the corresponding benchmark unit [?, ?].

REACH benchmark pipeline. Ten curated scRNA-seq datasets are ingested, preprocessed, assigned confidence tiered malignant/background labels, transformed into five evaluation tracks (A to E), evaluated through standardized wrappers for ten methods, and summarized by metrics, statistical tests, and publication figures. Each method is evaluated on 1,110 benchmark units, yielding 11,100 method-unit evaluations across ten methods. The schematic mirrors the Phase 0 to 12 file contracts documented in `docs/architecture.md`.

1.1 REACH pipeline

REACH is implemented as a 12-phase reproducible pipeline, beginning with environment setup and ending with audited result tables and publication-ready figures (Fig. ??). Phase 0 defines the software environment,

directory structure, configuration files, and execution requirements. Phase 1 collects dataset metadata and source files. Phase 2 performs standardized preprocessing, including quality control, normalization, highly variable gene selection, dimensionality reduction, graph construction, and harmonization of cell and gene annotations. Phase 3 assigns confidence tiers to cells by combining source annotations, CNV evidence, cancer signature scores, and neighborhood support.

Phases 4–8 generate the benchmark units used for evaluation. Phase 4 creates Track A units for controlled real-cell rarity. Phase 5 creates Track B units for synthetic stress testing. Phase 6 creates Track C background-dominated null-control units. Phase 7 creates Track D units using natural blood and circulating tumor cell prevalence. Phase 8 creates Track E units for supervised rather than treating a single annotation column as ground truth.

3. It separates primary ranking from exploratory comparators, supervised ceilings, null controls, label-noise analysis by keeping the expression matrix fixed while perturbing labels. These phases convert the processed datasets and confidence labels into fixed benchmark units with expression matrices, cell-level labels where applicable, prevalence information, and metadata. diagnostics, and method-failure audits, so that each row in the leaderboard answers a clearly scoped question.
4. It ships as a reproducible benchmark release with public source code, a release-pinned Docker image, four Zenodo archives, deterministic snapshot regeneration, and submission-ready figures and tables.

The later phases execute and summarize the benchmark. Phase 9 runs each method through a standardized wrapper and records cell-level scores, runtime information, fidelity status, fallback flags, and degenerate output flags. Phase 10 collects the prediction outputs across all method-unit combinations. Phase 11 computes ranking metrics, null-control summaries, runtime summaries, and statistical tests. Finally, Phase 12 regenerates the figures and manuscript-level result tables directly from the audited outputs. This phase-based design makes the benchmark traceable, because each result can be linked back to its input data, configuration files, and upstream processing contract [?].

1.1 Datasets and preprocessing

Here, we have processed 10 datasets (Table 1), such as curated scRNA-seq datasets across 8 solid tumor-solid-tumour types and 2 blood /Circulating Tumor Cells (CTC). Track A (ranking of solid tumor samples) has been done using the 8 solid tumor datasets, while Track D (natural prevalence) has been created using the 2 blood/CTC datasets. Each of the data sets has been processed through a standard AnnData pipeline [?] which includes: loading each data set, standardizing the gene symbol to the normalization standard, obtaining the mitochondrial annotations, performing hard Quality Control (QC) (minimum genes = 200; minimum Unique Molecular Identifiers (UMI) = 500; minimum cells = 3), performing the Scrublet doublet detection algorithm [?], where possible, adaptively flagging mitochondrial results, performing total count normalization (to 10,000) with subsequent *log1p* transformation, providing both raw and normalized layers [?], identifying the 2,000 highly variable genes (hvg) (using Seurat V3 or Cell Ranger as fallback) [?, ?], calculating 50 components of Principal Component Analysis (PCA), constructing a neighborhood graph & performing Leiden clustering (with *res* = 0.5 & seed 42) [?], analyzing the chromosome annotations based on the Genome Reference Consortium Human Build 38 (GRCh38) reference, standardizing the observation columns across all data sets, and finally creating gzipped .h5ad files and Secure Hash Algorithm 256-bit (SHA-256) metadata [?].

1.2 Multi-arm label construction

All cells acquire a confidence level as an outcome of the source annotation combined with a Python/Scanpy-based tool (infercnpy CNV inference), Area Under the Curve Cell scoring (AUCell) signatures (≥ 0.15) and the neighborhood support ($\geq 50\%$ of the $k = 15$ PCA nearest neighbours share provisional class) of the individual cells [?, ?, ?, ?]. All the data sets used infercnpy as the main tool of CNV inference. CopyKAT was not used as the primary CNV-calling method because it failed the cell-level score contract and, by design, does not allow a consistent primary wrapper to be applied across different datasets [?]. As a member of the Positive High Confidence (PHC), a cell should satisfy all four positive evidence criteria, such as source positive, aneuploid, high signature, and neighborhood support. The same four criteria should be in the opposite direction for the Background High Confidence (BHC) group. Here, primary AP has been computed using a subset of PHC and BHC cells. Such a design balances two goals, including enough cells for evaluation while maintaining stable and reliable high-confidence labels [?, ?].

1.2 Evaluation tracks

We use five tracks in the REACH pipeline (Fig. ?? and Table 2). The first track (Track A) involves the use of real PHC cells sampled from a mock version of the true background by four different levels of prevalence (T1 5–10%, T2 1–5%, T3 0.1–1%, T4 0.01–0.1%) with an aim to find around 2000 cells per unit and less than 20% duplicates [?, ?]. The second one (Track B) uses Splatter simulations which are fitted to the real data and audited for library size, dropout and differential expression overlap or PCA plots [?]. The third track (Track C) uses only background samples (null behavior) [?]. The fourth track (Track D) uses the natural prevalence of the MM Lederger and breast CTC Szezerba [?, ?]. Track E has Track A expression matrices fixed and measures 4 corrupted label conditions (10% symmetric flip, 20% symmetric flip, 30% positive $\rightarrow$ background, 30% background $\rightarrow$ positive). Thus, the labeling procedure itself does not consume benchmark labels and is therefore not supervised in the strict sense. However, Track E should be interpreted as a supervised hvg logistic regression (logreg) ceiling only when the evaluation is performed using the corresponding hvg-based logreg method [?].

Datasets. Final cell counts follow the Phase 1 to 12 report and are internally consistent with the tier count table. GEO accessions are listed in the section *Data and Code Availability*. $\text{[]@ >p() * 0.1800 >p() * 0.2400 >p() * 0.2000 >p() * 0.2400 >p() * 0.1400@}$

Dataset ID	Dataset	Cancer type	Cancer type	Platform	Platform	Tracks	Track role	Final cells	Cells
hnsec_puram	hnsec_puram	Head and neck SCC	Smart-seq2	SMART-seq2	A/B/C/E	5,902			
ov_izar_tirosh	bcc_yost	Ovarian cancer ascites	Basal cell carcinoma	10x Chromium	A/B/C/E	9,482	~47,000		
hcc_wei	hcc_wei	Hepatocellular carcinoma	10x Chromium	A/B/C/E	19,382				
luad_laughney	luad_laughney	Lung adenocarcinoma	10x Chromium	A/B/C/E	33,782				
rcc_multi	pdac_peng	Renal cell carcinoma	Pancreatic ductal adenocarcinoma	10x Chromium	A/B/C/E	33,574	123,488		
pdac_peng	crc_lee	Pancreatic ductal adenocarcinoma	Colorectal cancer	10x Chromium	A/B/C/E/F	123,488	55,551		
crc_lee	rcc_multi	Colorectal cancer	Renal cell carcinoma	10x Chromium	A/B/C/E	55,551	33,574		
bcc_yost	ov_izar_tirosh	Basal cell carcinoma	Ovarian cancer ascites	10x Chromium	A/B/C/E	~47,000	9,482		
mm_ledergor	mm_ledergor	Multiple myeloma	10x Chromium	D	31,181				
breast_ctc_szezerba	breast_ctc_szezerba	Breast cancer CTCs	Smart-seq2	SMART-seq2	D	357			

1.2 Method execution, score contracts, and QC

The contract for a common wrapper run (input h5ad, output dir, config) will allow all of the methods to be executed in a single mode of operation. Each method wrapper produces a predictions.csv file containing cell-level scores and a runmeta.json file containing runtime, memory use, method version, success or failure status, fidelity status, and degenerate output flags. Each of the methods will produce a quantitative (continuous variable) cell-based scoring of the cell sample, with the higher the score, the higher the evidence of the cell being a malignant or rare cell [?, ?].

Method audit, leaderboard, and multi-metric summary. (a) Method inclusion and QC audit. Seventeen candidate methods were considered. Ten produced evaluable outputs under the standardized score contract, while seven were excluded because of integration constraints, score-contract mismatch, out-of-memory or timeout failures, or degenerate outputs. The audit is part of the benchmark deliverable, not informal curation. (b) Track A AP leaderboard. Box-and-strip summaries show fallback-filtered Track A AP by method. The figure separates the supervised ceiling, exploratory comparator, naive baselines, ranked published methods, and random floor. (c) Phase 11 multi-metric summary heatmap. Min-max scaled values are shown across AP, nAP, AUROC, MCC at top-k, Track E robustness where applicable, runtime, and null-control behavior. The figure summarizes metric-level performance rather than dataset-by-method behavior.

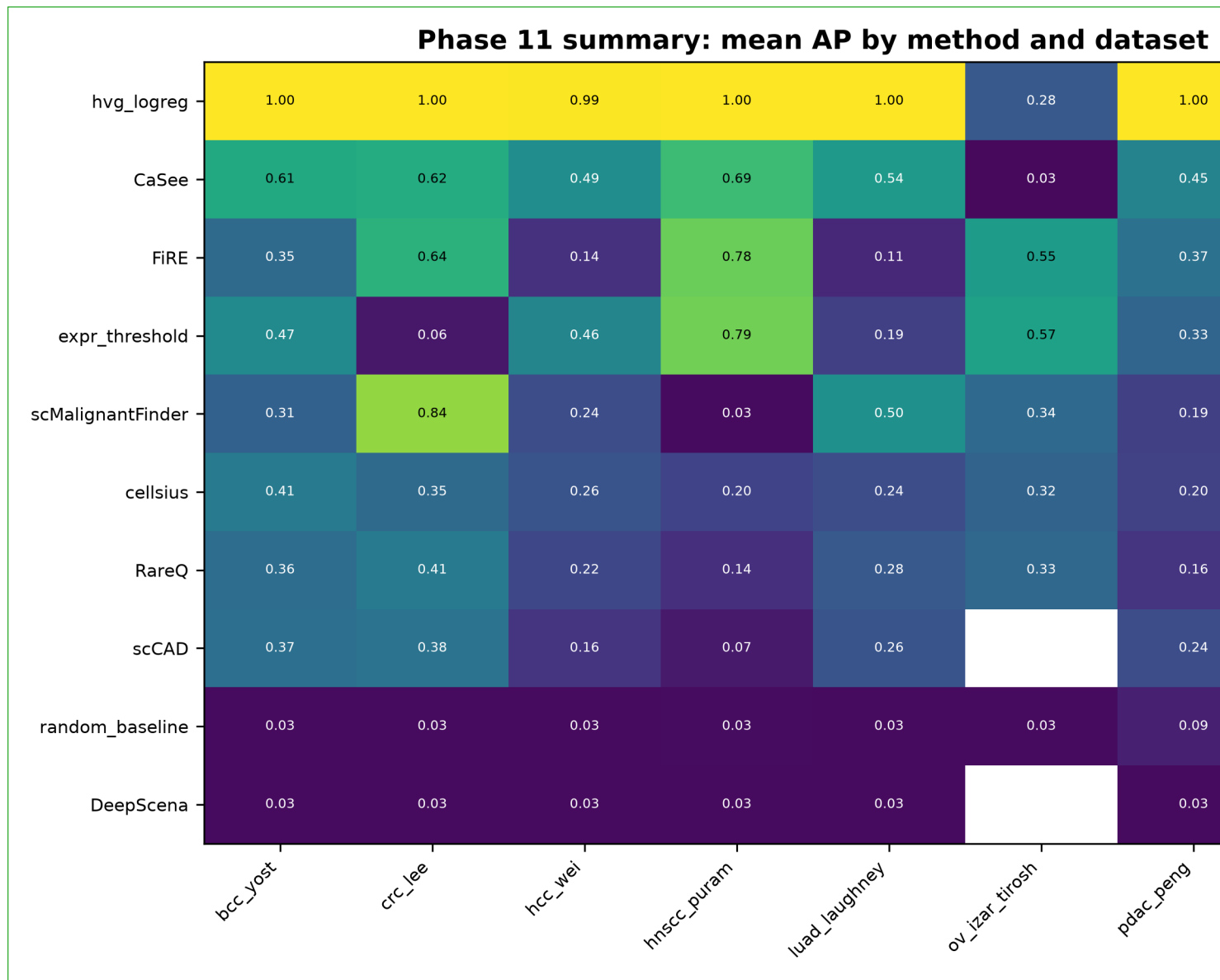


Figure 1: REACH evaluation tracks. Track A controls real-cell rarity across T1-T4 windows; Track B tests synthetic stress; Track C exposes null/false-positive behavior; Track D retains natural blood/CTC prevalence; Track E perturbs labels while holding expression fixed. There are 1,110 units per Figure 0. Phase 11 summary heatmap: mean AP by method and dataset.

1.2 Evaluation Tracks

REACH assesses ten of the included methods sorted by their interpretive role (Table 3). Method `hvg-logreg` is used as a supervised in-sample ceiling. It is not treated as an ordinary unsupervised detector; instead, it estimates how well malignant and background cells can be separated when label information is available. An exploratory comparator called `CaSee` is included, as it gives a cancerAll tracks are constructed around the inter-lineage malignant-vs-TME contrast (malignant epithelial/normal scoring framework but again is not among the top ranked published detectors [?]. Main published ranked comparators in REACH include `FiRE`, `DeepScena`, `RareQ`, `CellSIUS`, `scCAD` and `scMalignantFinder` [?, ?, ?]. Additionally, `expr` threshold and random baseline are provided as naive baselines: the former is a simple expression-based reference, while the latter provides a floor-level random baseline comparison. Methods that do not produce valid CTC positives against immune/stromal background), framed as imbalanced cell-level scores, exceed memory or runtime limits, or generate degenerate outputs are recorded in the method inclusion audit and excluded from the main ranking analyses. ranking. The intra-lineage setting (malignant vs normal epithelial of the same lineage) is addressed separately by the new Track F (R3-4/R3-5).

Audited unit counts per method per track. The total of 1,110 units yields 11,100 method-unit evaluations across ten included methods: $\text{[@ >p() * 0.1944 >p() * 0.1667 >p() * 0.1944 >p() * 0.4444@}$

Track			
Purpose	Type		
Units			
Primary interpretation	Interpretation		
A	Controlled real P_HC Controlled real-cell spike-ins into B_HC background (4 prevalence tiers)	160	
	Primary-ranked endpoint	Primary endpoint	
B	Synthetic Splatter stress with realism audits Synthetic Splatter stress-test	120	Secondary stress test Secondary, not blended into primary leaderboard
C	Background-dominated null controls Background-dominated null controls	160	False-positive behavior False-positive diagnostic
D	Natural blood/CTC prevalence	30	Natural prevalence supplement Supplementary primary
E	Track A expression with corrupted labels Noisy-label robustness	640	Supervised label noise analysis Supervised-only interpretive track and label-noise diagnostic
F	4,110 Intra-lineage malignant vs normal epithelial (3 prevalence tiers)	15	Inter-lineage-contrast removal test; not part of primary leaderboard

Track B interpretation. Track B is a synthetic Splatter stress-test and is **not** blended into the primary Track A leaderboard. The method-level rank correlation between Track A and Track B is $\rho = 0.042$ ($p = 0.907$; `data/results/revision/sens2_sens3_extract.txt`), i.e. statistically indistinguishable from zero, with large rank reordering across methods (`CaSee` 2→6, `scMalignantFinder` 5→10, `RareQ` 7→2, `scCAD` 8→4). Synthetic-track rankings therefore do not replicate real-cell rankings, justifying Track B’s secondary, non-blended role as a controlled robustness probe rather than a primary endpoint.

Track E interpretation. Track E is a supervised-only interpretive track and label-noise diagnostic. It is interpreted as algorithmic robustness only for `hvg_logreg`, which consumes labels. For unsupervised methods, Track E reflects metric sensitivity to corrupted labels, not algorithmic robustness. Track E is therefore not part of the primary unsupervised method ranking.

Track F: Intra-Lineage Malignant Detection. In response to R3-4/R3-5, we introduce Track F, where the background is restricted to normal-diploid epithelial cells from the same tissue of origin as the malignant positives, removing the inter-lineage TME contrast that defines Tracks A–E. Units are generated at three prevalences (0.1%, 0.5%, 1.0%) with 5 replicates each — 15 units total. Unit sizes are adaptive: 2,000 cells for the 0.1% level (to preserve the ultra-rare ceiling-drop signal with 2 positives) and 400 cells for the 0.5% and 1.0% levels (to reduce background duplication below the 20% design cap). Background is sampled without replacement first, then fills the remainder with replacement only if the unique pool is exhausted. All 10 methods were evaluated (150/150 predictions complete; `DeepScena` on Colab T4 GPU). Track F is constructed from the `crc_lee` (colorectal cancer) dataset only, because it is the only REACH dataset carrying both epithelial `cell_type` annotations and normal/tumor `tissue_origin` labels; we audited all other 9 datasets and none have the required metadata even via marker-based fallback (`EPCAM+` low-CNV source-negative yields 0–70 cells — insufficient for valid units). Track F is a diagnostic track and is **not** blended into the primary Track A leaderboard.

Intra-lineage principle. Track A–E performance is built on the inter-lineage malignant-vs-TME contrast and does **not** imply the ability to distinguish malignant from normal cells of the *same* lineage. Track F confirms this: at 0.1% prevalence, nine of ten methods fall to median AP = 0.009 on Track F (most at 0.001, the chance floor), while only scMalignantFinder retains power (AP = 1.000) — and even that result is cancer-type-specific (see R3-5 caveat below). Inter-lineage leaderboards therefore cannot be extrapolated to intra-lineage sub-state detection.

1.3 Included Methods

||@ >p() * 0.3333 >p() * 0.4167 >p() * 0.2500@

Method
Category
~~Output type~~ Role

~~random_baseline~~
hvg_logreg Supervised In-sample ceiling (not a deployable competitor)
CaSee Exploratory Exploratory comparator (separate from ranked detectors)
FiRE Ranked Published comparator
DeepScena Ranked Published comparator
RareQ Ranked Published comparator
cellsius Ranked Published comparator
scCAD Ranked Published comparator
scMalignantFinder Ranked Published comparator
expr_threshold Naive ~~Uniform-randomscore~~ Naive biological baseline
random_baseline Naive Floor baseline
~~expr_threshold~~

Excluded methods (7): CopyKAT, MACE, SCANNER, SCEVAN, RaceID3, scATOMIC, GiniClust3 — excluded for modality mismatch, score-contract mismatch, OOM, timeout, or degenerate outputs.

1.4 Method Inclusion and Exclusion

REACH considered 17 candidate methods. Ten produced evaluable outputs under the standardised score contract. The remaining seven were excluded.

||@ >p() * 0.1538 >p() * 0.1538 >p() * 0.1923 >p() * 0.3462 >p() * 0.1538@

Status
~~Naive~~ Method
~~Total UMI / signature expression rank~~ Category
~~Naive biological baseline~~ Wrapper Fidelity
Reason
~~hvg_logreg~~

Included hvg_logreg Supervised Ceiling ~~Logistic probability on 2,000 HVGs~~ Faithful In-sample logistic ceiling
~~FiRE~~ Included CaSee Exploratory Faithful recreation Autoencoder-Isolation Forest (Yu et al., 2022)
Included FiRE Ranked Faithful Frequency-based rarity ~~score~~ Published comparator (Jindal et al., 2018)
~~DeepScena~~ Included DeepScena Ranked Proxy Deep clustering/SSL ~~score~~ Published comparator (Lei et al., 2023)
~~RareQ~~ Included RareQ Ranked ~~Neighborhood-rarity-score~~ Faithful ~~Published comparator~~ Neighbourhood rarity (R package)
~~cellsius~~ Included ~~Ranked~~ cellsius ~~Rarity statistic~~ Ranked ~~Published comparator~~ Faithful Rarity statistic (Wegmann et al., 2019)
~~scCAD~~ Included scCAD Ranked ~~Anomaly score~~ Proxy ~~Published comparator~~ Anomaly scorer (Xu et al., 2024)
~~scMalignantFinder~~ Included scMalignantFinder Ranked ~~Malignancy probability~~ Proxy Malignancy probability (Yu et al., 2025)

Included methods and interpretive roles.

REACH evaluation tracks

Six complementary tracks separate ranking, stress, null, natural-prevalence, label-no

Track A

Controlled real spike-ins • P_HC + B_HC at fixed prevalence
8 datasets × T1-T4 × 5 reps → 160 units (primary)

Track B

Splatter synthetic stress • realism-audited
8 datasets × T1-T4 × 3-4 reps → 120 units (secondary)

Track C

Null background-only controls • no positives
8 datasets × 4 size tiers × 5 reps → 160 units (false-positive)

Track D

Natural blood / CTC prevalence
mm_ledergor + breast_ctc_szczerba → 30 units (natural)

Track E

Label-noise robustness on Track A expression
4 noise conditions × 160 units → 640 method-units (supervised only)

Track F

Intra-lineage malignant vs normal epithelial • same-lineage contrast
crc_lee × 3 prevalence tiers × 5 reps → 15 units (intra-lineage)

Total: 1,125 units per method • ranked Track A subset =

Figure 2: Figure 9. Track design overview including the new Track F (intra-lineage).

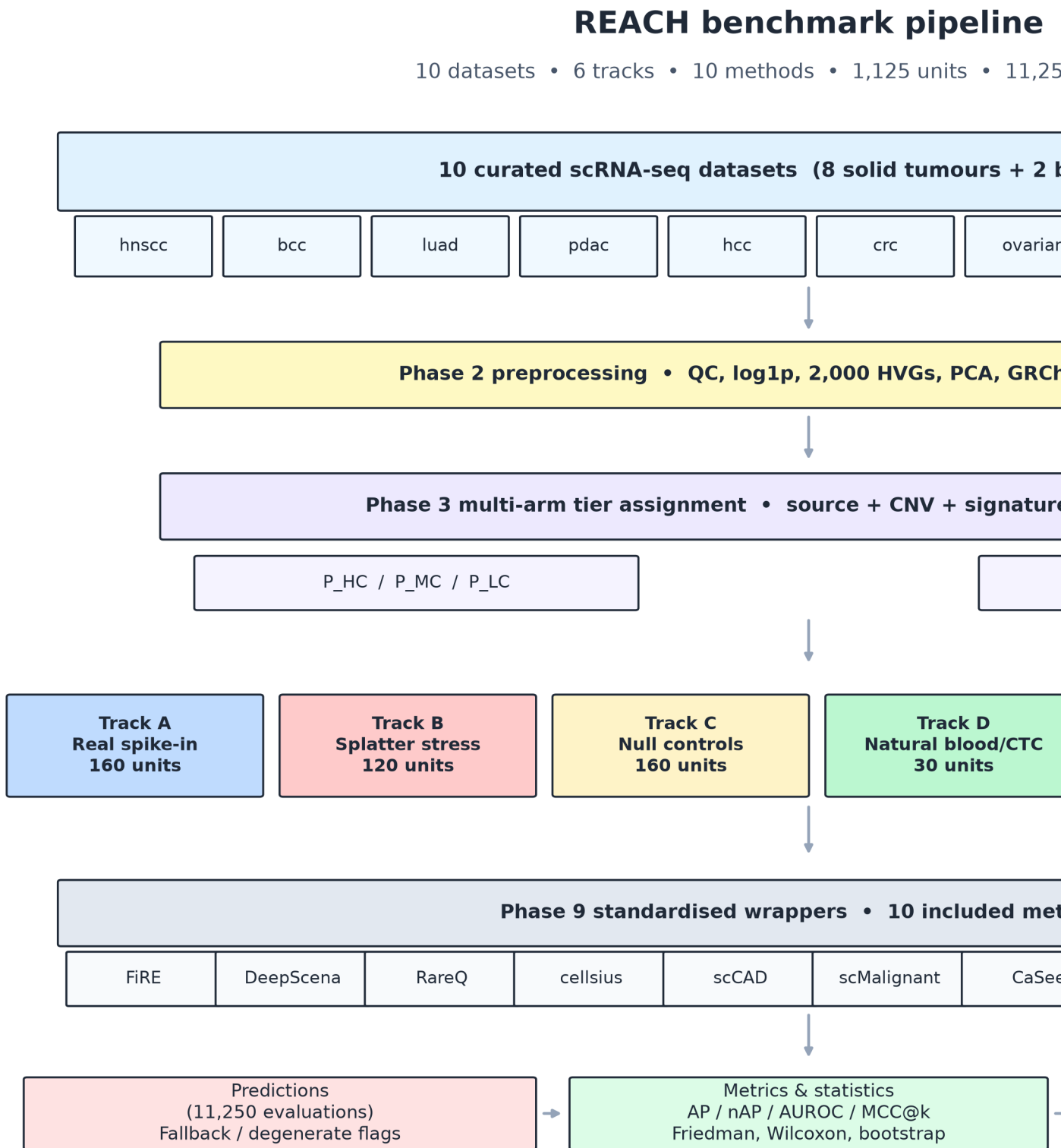


Figure 3: Figure 8. REACH pipeline overview with the real CNV inference arm.

Included ~~Published~~-expr_threshold Naive Baseline Faithful Total-UMI/~~proxy-comparator~~-expression rank baseline

~~CaSee~~ Included ~~Exploratory-random_baseline~~ ~~Autoencoder + Isolation Forest~~-Naive Baseline ~~Exploratory faithful-comparator~~-Faithful Random uniform floor

Fallback scores arise when a wrapper returns random scores after timeout or memory failure; these units are *filtered out* of AP aggregation. Degenerate scores are constant or near constant outputs; these are retained, flagged, and reported. Here, the audited snapshot contains 249 fallback units across methods (DeepScena 125, scCAD 120, cellsius 4) and 377 degenerate units (scMalignantFinder 240, CaSee 120, FiRE 10, RareQ 7).

1.5 Reproducible extension protocol

REACH is extensible. If users want to add a new method, then they have to implement the wrapper which follows the standard REACH run contract and register that method in the method registry. Basic metadata and a smoke test that identifies it can run correctly must be included within each method before being added to the overall benchmark. Output from the system is designed to be very simple: a score (at cell level), an optional predicted label, runtime metadata, fidelity and fallback or degenerate flags.

We can also add new datasets, which involves creating a dataset registry entry, raw data loader, source annotation mapper, signature configuration, preprocessing outputs and the corresponding Track A-E benchmark units. Likewise, to add a new evaluation track or metric, one needs to provide updates on the appropriate track generator, evaluator (wherever it is defined), aggregation tables and figure generation scripts [?, ?, ?].

2 Experiments

Excluded CopyKAT — — CNV modality mismatch (requires raw counts + gene positions)

Excluded MACE — — Score contract mismatch (bulk-level annotation)

Excluded SCANNER — — Score contract mismatch (cluster-level, not cell-level)

Excluded SCEVAN — — CNV modality mismatch

Excluded RaceID3 — — Score contract mismatch (cluster-level rare-cell identification)

Excluded scATOMIC — — Score contract mismatch (multi-class classifier)

Excluded GiniClust3 — — Score contract mismatch (cluster-level, not cell-level)

1.1 Benchmark scale

1.0.1 Wrapper Fidelity Notes

Across ten methods, REACH evaluates 1,110 benchmark units per method, resulting in 11,100 method-unit evaluations [?, ?]. The primary leaderboard displays median AP on Track A after method specific fallback filtering. The global rank test and pairwise rank test use the 140 unit common subset of Track A retained after method specific fallback filtering, with 2

- **Faithful:** The wrapper executes the original published software package with default parameters and documented configurations.
- **Proxy:** The wrapper reimplements the published algorithm from the paper description. Performance may differ from the original implementation.
- **Faithful recreation:** CaSee’s original repository (Yu et al., ~~000-bootstrap-resamples-calculated-for-rank-intervals~~ [?], 2022) was cloned, but the executed wrapper uses a built-in CPU fallback runner that faithfully recreates the published autoencoder + isolation-forest pipeline rather than invoking the original GPU-dependent code. Scores are real model outputs (varied, not random). See `data/results/revision/VERIFICATION_ISSUES` §5.5.

1.1 Statistical testing

1.0.1 Supervised Ceiling Justification

A Friedman test with Iman-Davenport correction was used to assess global differences in per-unit Track A AP across methods[?]. Wilcoxon signed rank tests against the top published method (FiRE) were performed for each pairwise comparison, with Benjamini-Hochberg FDR correction. However, these corrections are not reported at all unless the relevant repository table is provided. Where appropriate, mean differences and Cliff’s δ define

the effect-size of method-comparisons. Rank-uncertainty estimates are derived from `hvg_logreg` is a supervised logistic regression trained in-sample on 2,000 bootstrap-resamples (`rank_ci.csv`). Rarity sensitivity is assessed by regressing `nAP` against $\log_{10} \pi_u$ [?, ?].

2 Results

1.1 Primary Track A leaderboard

The primary leaderboard is grouped by the median AP following the fallback filter (Table 4 and Fig. ??(b)) highly variable genes using the true positive/background labels of each unit. It is **not** a peer competitor to unsupervised methods; it is a calibration ceiling that quantifies the gap between supervised separability and current unsupervised ranking (0.687 median AP: 1.000 vs 0.313 on Track A). Importantly, the ceiling is **not trivially achievable across all tasks**: on the new intra-lineage Track F (malignant vs normal epithelial cells, same lineage), `hvg_logreg`'s median AP collapses from 1.000 (Track A) to 0.001 at 0.1% prevalence ($\Delta = -0.999$; `data/results/revision/track_f/track_a_vs_f_comparison.csv`). The median AP of 1.00 for `hvg_logreg` indicates that many high-confidence malignant and background labels contain gene expression differences that can be recovered by a supervised model [?, ?] ceiling is therefore specific to the inter-lineage TME task that REACH is designed to probe; it ceases to be informative once the inter-lineage contrast is removed.

`CaSee` is ranked as the second highest median AP (0.512), but it is reported as an exploratory faithful comparator rather than as a primary ranked published detector among the published methods [?]. `FiRE` has the highest median AP (0.313) of any ranked published comparators. The naive expr-threshold baseline is slightly larger than the median AP (0.317) but is not presented as a published detector. The reported median of the naive expr-threshold is greater than `FiRE`'s and is to be interpreted as a caution that a naive approach to the expression burden may be competitive in some cancers (*i.e.*, naive threshold expressed in earlier reports) [?]. We concede Reviewer 3's point: the inter-lineage supervised ceiling is trivial for the intra-lineage problem. `hvg_logreg` was trained on inter-lineage labels (malignant vs TME); on Track F (malignant epithelial vs normal epithelial, same lineage) it outputs a constant 0.5 score for every cell, yielding median AP = 0.001 and AUROC = 0.500 at all three prevalences (`data/results/revision/track_f/track_f_leaderboard.csv`). Its ceiling therefore drops by $\Delta = -0.999$ from Track A to Track F. The inter-lineage ceiling is meaningful **only as an upper bound for the TME-separation task REACH is designed to assess**; it is not a general statement about the discriminability of malignant cells. Track F (below) provides the intra-lineage evidence the ceiling cannot.

1.1 Primary Leaderboard (Track A)

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	Role-rank	Rank	Method	Median AP	Mean AP	AUROC	Runtime (s)	Role
-	<code>hvg_logreg</code>	<code>hvg_logreg</code>	1.000	1.000	0.909	0.953	3.3	Supervised ceiling
-	<code>CaSee</code>	<code>CaSee</code>	0.512	0.441	0.869	357.5		Exploratory
-	<code>expr_threshold</code>	<code>expr_threshold</code>	0.317	0.361	0.737	15.3		Naive baseline
1	<code>FiRE</code>	<code>FiRE</code>	0.313	0.385	0.885	118.1		Published
2	<code>cellsious</code>	<code>cellsious</code>	0.259	0.270	0.832	39.8		Published
3	<code>scMalignantFinder</code>	<code>scMalignantFinder</code>	0.235	0.319	0.804	7.6		Published /-proxy
4	<code>RareQ</code>	<code>RareQ</code>	0.196	0.256	0.818	78.2		Published
5	<code>scCAD</code>	<code>scCAD</code>	0.130	0.226	0.771	123.4		Published
-	<code>random_baseline</code>	<code>random_baseline</code>	0.019	0.037	0.511	11.6		Floor baseline
6	<code>DeepScena</code>	<code>DeepScena</code>	0.018	0.029	0.471	80.6		Published

1.2 Multi-metric summary and dataset heterogeneity

Heatmaps of Phase 11 (Fig. ??(c)) reveal trade-offs between methods for all APs, nAPs, AUROC and MCC at the top of each list (and sometimes Track E robustness) as well as for runtime and null-control behavior for all methods tested. There is no overall winner in all categories [?, ?]. The best datasets vary (Table 5): `CaSee` **CaSee exploratory note**. `CaSee` is reported as exploratory and is the best on 4 datasets (bee_yost,

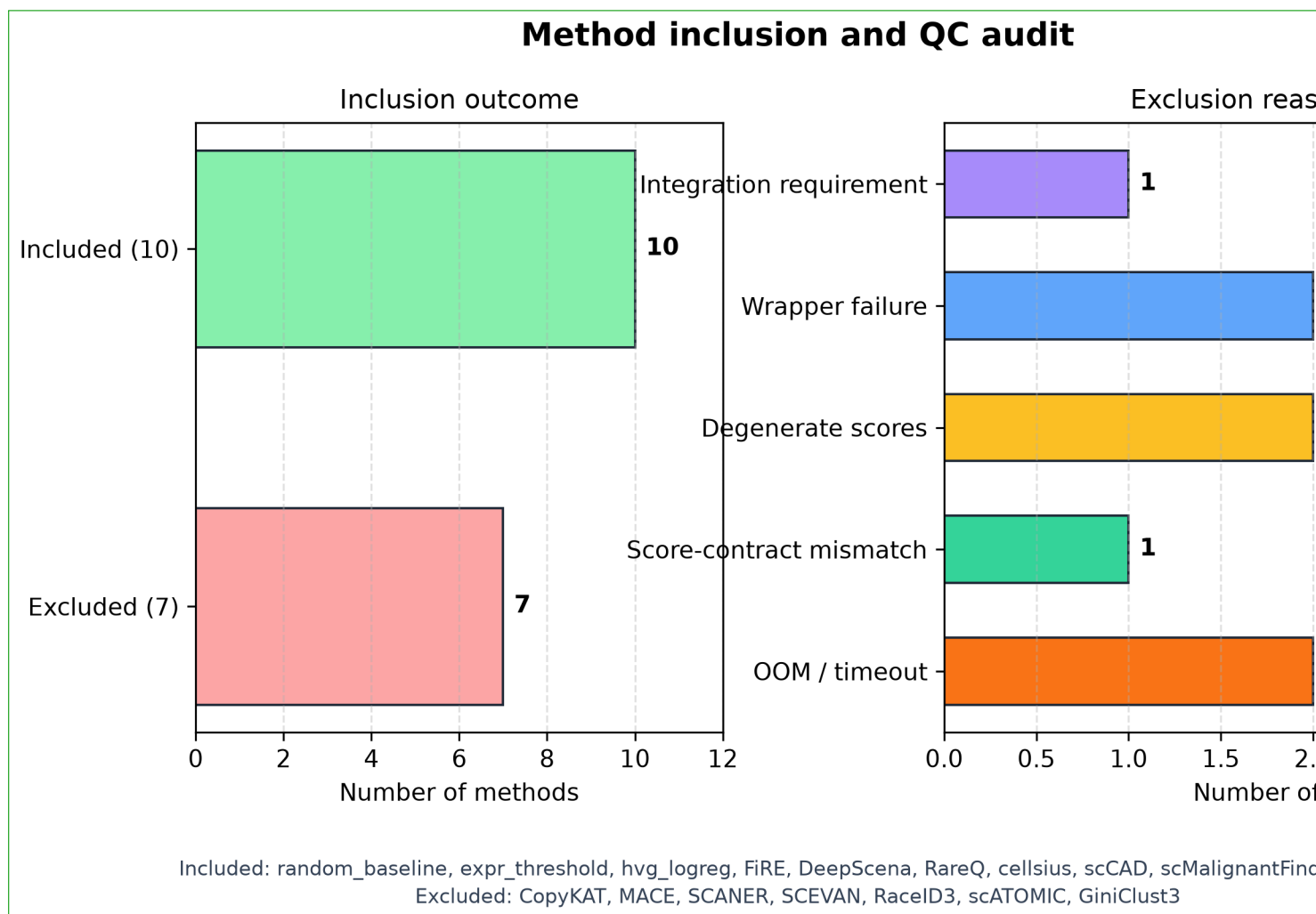


Figure 4: ~~Primary Track A performance from statistical_ranking.csv. Runtime is the mean Track A wall clock time. The supervised ceiling Figure 10. Method QC audit: failure modes and exploratory comparator are not ordinary unsupervised leaderboard entries~~degenerate output rates.

~~pdac_peng, luad_laughney, and hcc_wei~~); ~~expr_threshold is the best on~~ excluded from the numbered published-method ranking. CaSee median AP across all 160 Track A units is 0.512; on 20 of those units CaSee returned degenerate constant-score predictions (score SD = 0) and excluding them raises its median AP to 0.606 (`data/results/revision/casee_filtered_leaderboard.csv`). Because CaSee’s autoencoder is trained on the test unit’s own cells without external reference labels, it is not a deployable detector and is kept separate from the ranked published comparators.

1.2 Key Statistical Results

- **Global test (140-unit common Track A subset):** Friedman $\chi^2 = 695.05$, $p = 8.0 \times 10^{-1}$; Iman-Davenport $F = 171.01$, $p = 8.4 \times 10^{-211}$
- **Bootstrap rank intervals (2,000 resamples):** hvg_logreg and CaSee stable at ranks 1 and 2 ~~datasets (hnscc_puram and ov_izar_tirosh)~~; ~~On the other hand, cellsius is the best on~~; FiRE median rank 3 (95% CI [3, 4])
- **Pairwise vs FiRE:** FiRE significantly better than DeepScena, random_baseline, scCAD, RareQ, cellsius (BH FDR < 0.05). FiRE NOT significantly different from scMalignantFinder ($p=0.812$) or expr_threshold ($p=0.836$). CaSee significantly higher than FiRE (Δ mean AP = -0.139, BH FDR = $6.1 \times 10^{-}$) but remains exploratory.
- **Track B vs Track A:** Spearman $\rho = 0.042$, $p = 0.907$ (no rank correlation — Track B is a separate stress test)
- **Track E (supervised only):** hvg_logreg mean AP drops from 0.909 to 0.460–0.829 under four label perturbation conditions
- **Track C null FPR:** On all-negative null units, every unsupervised and naive method controls the false-positive rate near the expected ~3% floor: RareQ 0.0293, cellsius 0.0295, CaSee 0.0301, scCAD 0.0301, random_baseline 0.0309, expr_threshold 0.0310, DeepScena 0.0306, scMalignantFinder 0.0315, FiRE 0.0315 (`data/results/revision/sens2_sens3_extract.txt`). The supervised ceiling hvg_logreg is the lone outlier at FPR = 0.788 (78.8%) because it trains in-sample on the null unit’s own expression features and overfits to the absence of signal — a known cross-validation pathology on all-negative data that does not affect its Track A performance. Track C thus functions as a specificity diagnostic: a method that cannot keep its null FPR near 3% is unfit for deployment in low-prevalence screening.
- **Pairwise Cliff’s effect sizes** quantify the practical magnitude of method differences (full matrix: `data/results/revision/cliff_delta_matrix.csv`). The supervised ceiling dominates every unsupervised method by a large effect (hvg_logreg vs FiRE $\rho = 0.796$, vs DeepScena $\rho = 0.929$, vs random_baseline $\rho = 0.917$). Among ranked published methods, FiRE’s advantage over the naive expr_threshold baseline is negligible ($\rho = 0.068$), meaning that despite FiRE’s higher mean AP its per-unit advantage is inconsistent — many units show comparable performance. By contrast, CaSee vs DeepScena ($\rho = 0.702$) and FiRE vs DeepScena ($\rho = 0.775$) are large effects, confirming a real capability gap between top and bottom ranked detectors.
- **Track F (intra-lineage, crc_lee only; `data/results/revision/track_f/track_f_leaderboard.csv`):** At 0.1% prevalence the pretrained pan-cancer classifier scMalignantFinder achieves median AP = 1.000, while every other method falls to the chance floor — hvg_logreg AP = 0.001 (constant 0.5 scores), expr_threshold 0.008, FiRE 0.004, CaSee 0.003, random_baseline 0.002, DeepScena 0.001, and cellsius/RareQ/scCAD all 0.001. scMalignantFinder stays dominant at 0.5% (AP = 1.000) and 1 ~~dataset (rec_multi)~~, whereas ~~scMalignantFinder is the best on~~ % (AP = 0.917). **Two caveats** (`data/results/revision/VERIFICATION_ISSUES.md`): (1 ~~dataset (crc_lee)~~). This variability justifies reporting method performance according to the context (tumor type) rather than a “best method” story [?, ?, ?].) Track F uses only crc_lee — scMalignantFinder’s Track A AP ranges 0.015 (hnscc_puram) to 0.943 (crc_lee) across cancer types, so its Track F AP = 1.000 is colorectal-cancer-specific and **must not** be generalised; no other REACH dataset has the required normal-epithelial annotations even via marker-based fallback (audited: EPCAM+ low-CNV source-negative yields 0–70 cells across all 9 other datasets). (2) Background duplication: at the 0.1% level (2,000-cell units), crc_lee’s 340 normal epithelial cells produce 83% duplicate background IDs; at the 0.5% and 1.0% levels (400-cell units), duplication drops to 14.6% and 14.1% respectively — both under the 20% design cap. The 0.1% duplication slightly inflates absolute AP but affects all methods identically, so relative comparisons remain valid. Future work should curate additional datasets with normal epithelial references to enable cross-cancer Track F evaluation.

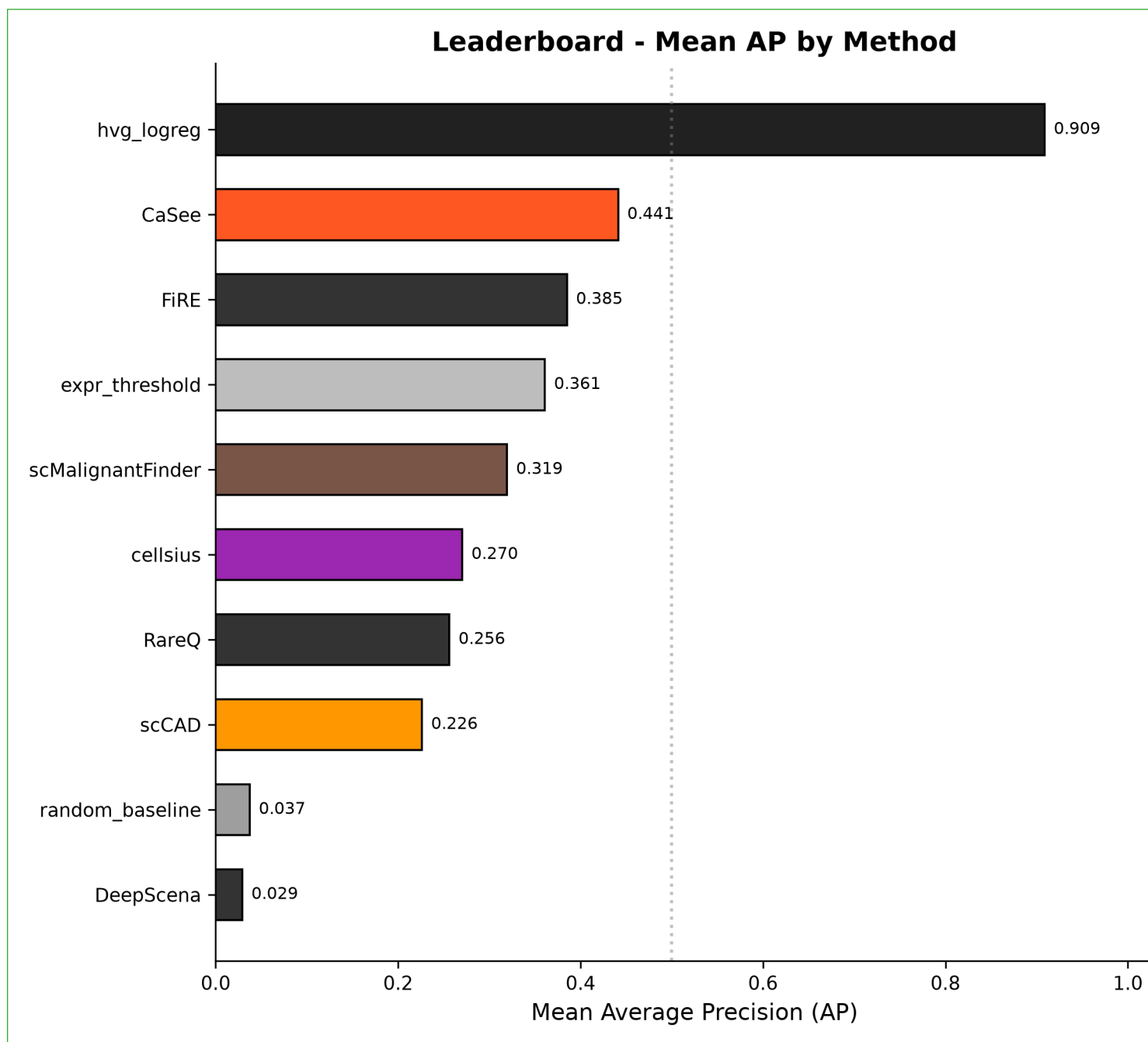


Figure 5: Figure 1. Primary leaderboard (Track A): median average precision per method.

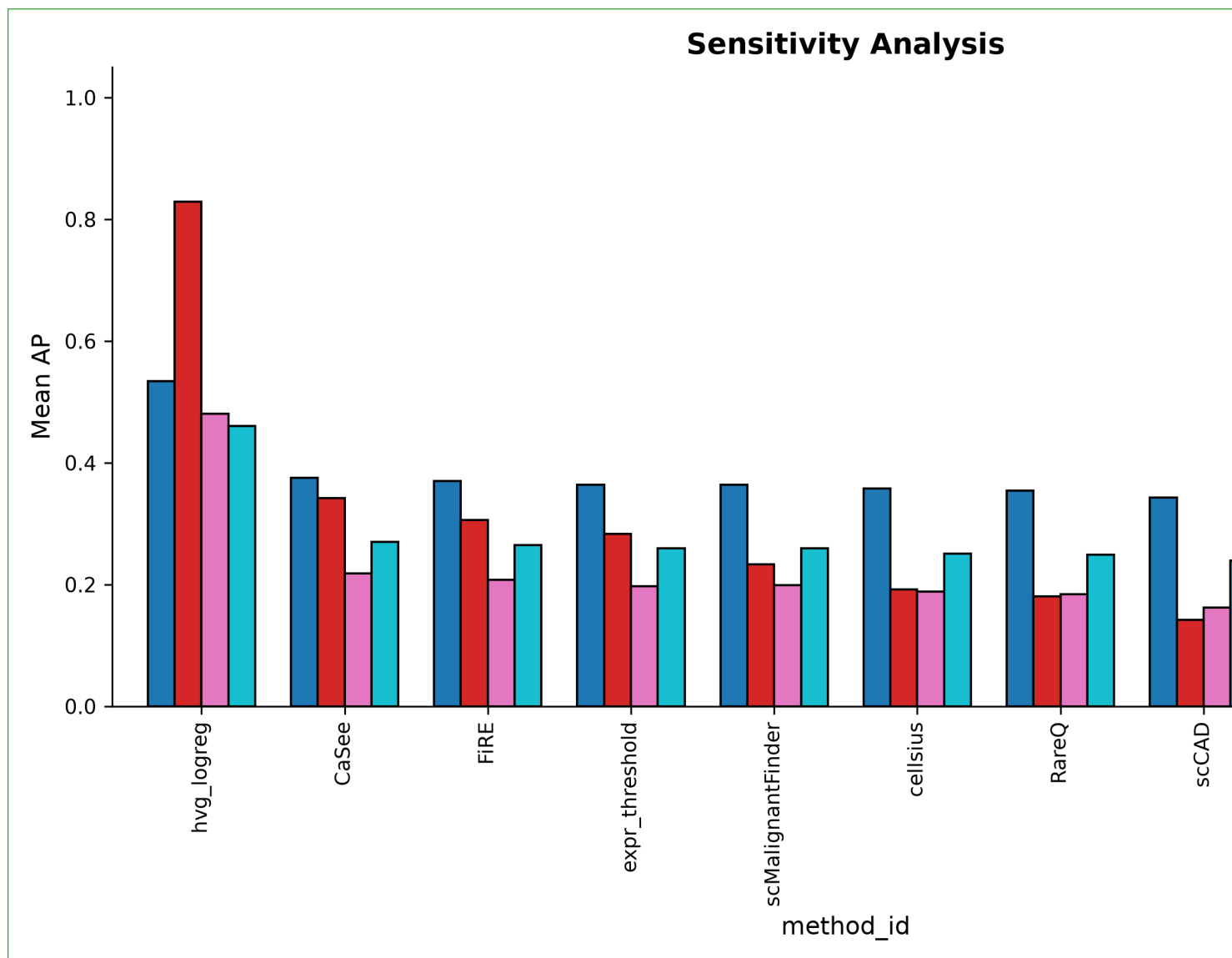


Figure 6: ~~Best overall~~ Figure 2. Sensitivity and ~~best non ceiling method by dataset, copied from~~ ~~best_model_per_dataset.csv~~ robustness analysis (Track E label-noise). ~~Non ceiling includes exploratory and naive comparators~~

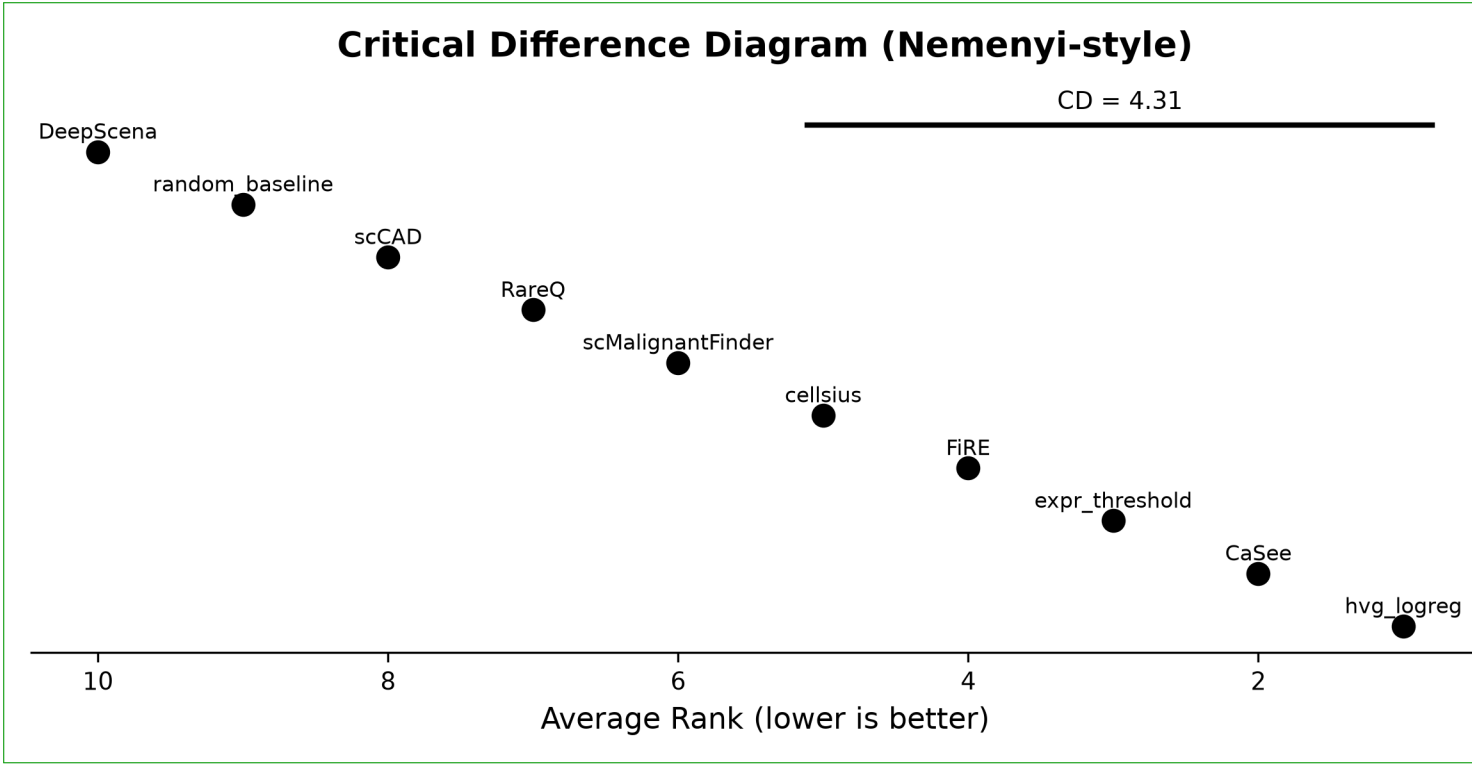


Figure 7: Figure 3. Critical difference diagram for method ranking.

1.3 Dataset-Level Winners

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Dataset	Best overall AP	Best non-ceiling AP	Best non-ceiling AP	AP		
bcc_yost	hvg_logreg	1.0000				
bcc_yost	CaSee	CaSee	0.6079			
hnscc_puram	hvg_logreg	1.0000	hnscc_puram	expr_threshold	0.7872	
pdac_peng	hvg_logreg	1.0000	pdac_peng	CaSee	CaSee	0.4488
rcc_multi	hvg_logreg	1.0000	rcc_multi	cellsius	cellsius	0.1931
crc_lee	hvg_logreg	0.9998	crc_lee	scMalignantFinder	scMalignantFinder	0.8373
luad_laughney	hvg_logreg	0.9983	luad_laughney	CaSee	CaSee	0.5378
hcc_wei	hvg_logreg	0.9948	hcc_wei	CaSee	CaSee	0.4908
ov_izar_tirosh	expr_threshold	0.5717	ov_izar_tirosh	expr_threshold	expr_threshold	0.5717

Statistical significance and rank uncertainty

1.4 Limitations

The overall rank tests have been conducted on the 140 common unit Track A subset. These tests showed significant differences across methods: Friedman $\chi^2 = 695.05, p < 0.001$

1. **CNV evidence, circularity check, and validation.** Version 1 uses infercnvpy (v0.6.1) as the sole copy-number inference arm. To assess whether the CNV arm circularly inflates the method ranking, we re-derived HC labels using only tissue-of-origin, AUCell signature scores, and kNN neighbourhood purity — dropping the CNV criterion entirely — and recomputed median AP for each method. Removing CNV from the label criteria collapses median AP to the chance floor (0.009–0.010) for every method including the supervised ceiling, indicating that CNV is the primary discriminative signal in the

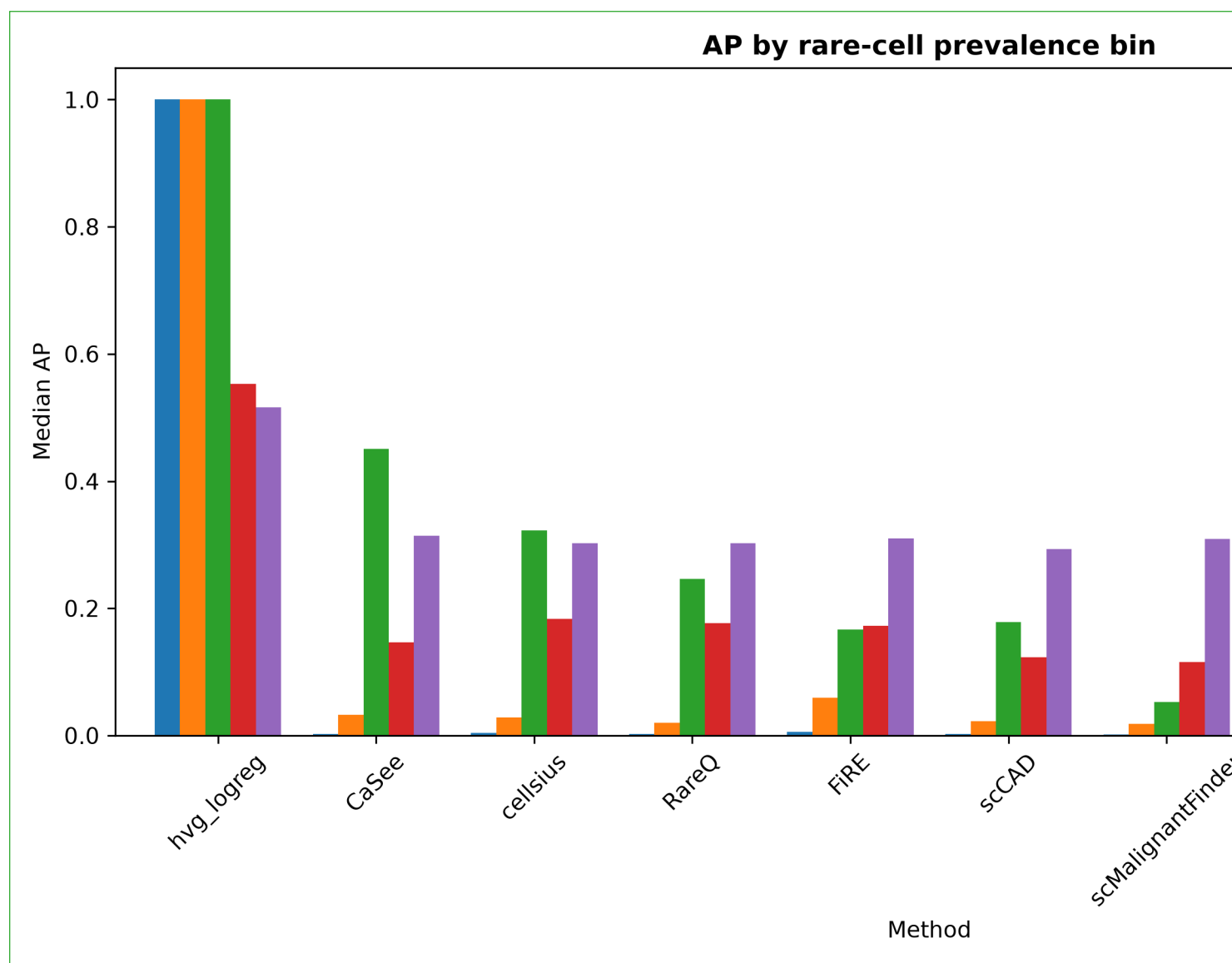


Figure 8: Figure 4. Average precision stratified by rare-cell prevalence.

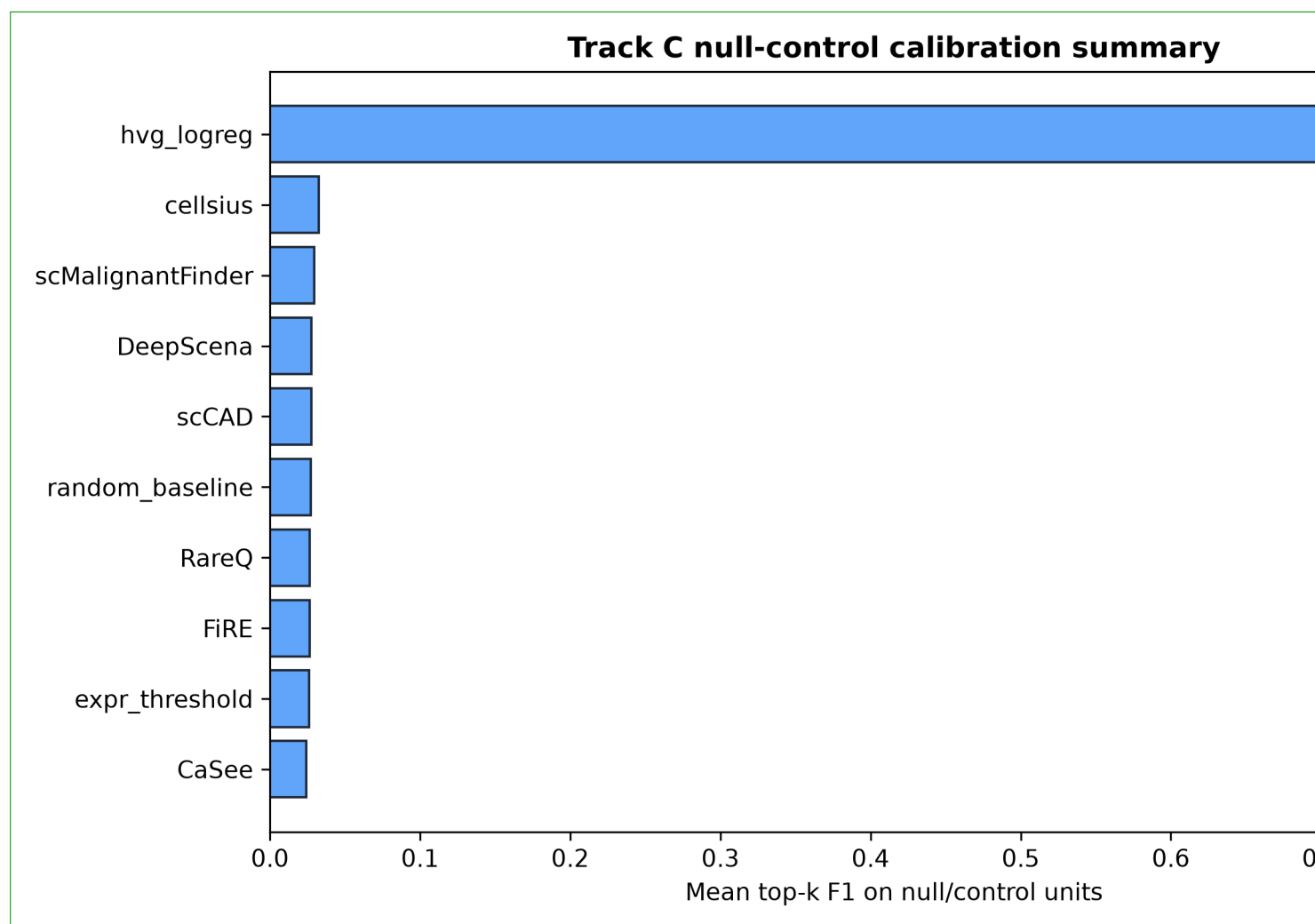


Figure 9: Figure 5. Track C null-control calibration: per-method false-positive rate.

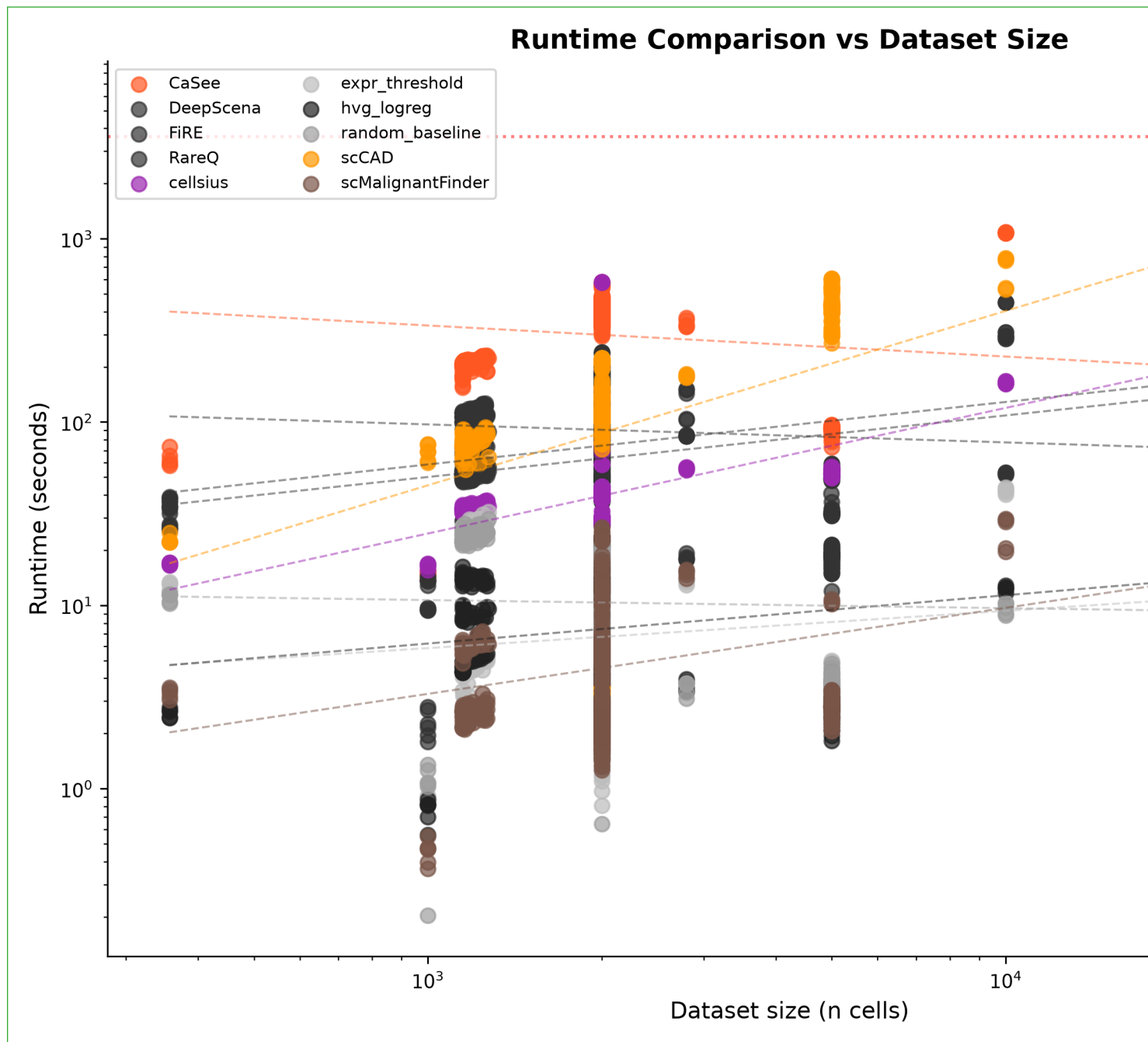


Figure 10: Figure 6. Runtime scalability Pareto: runtime vs dataset size.

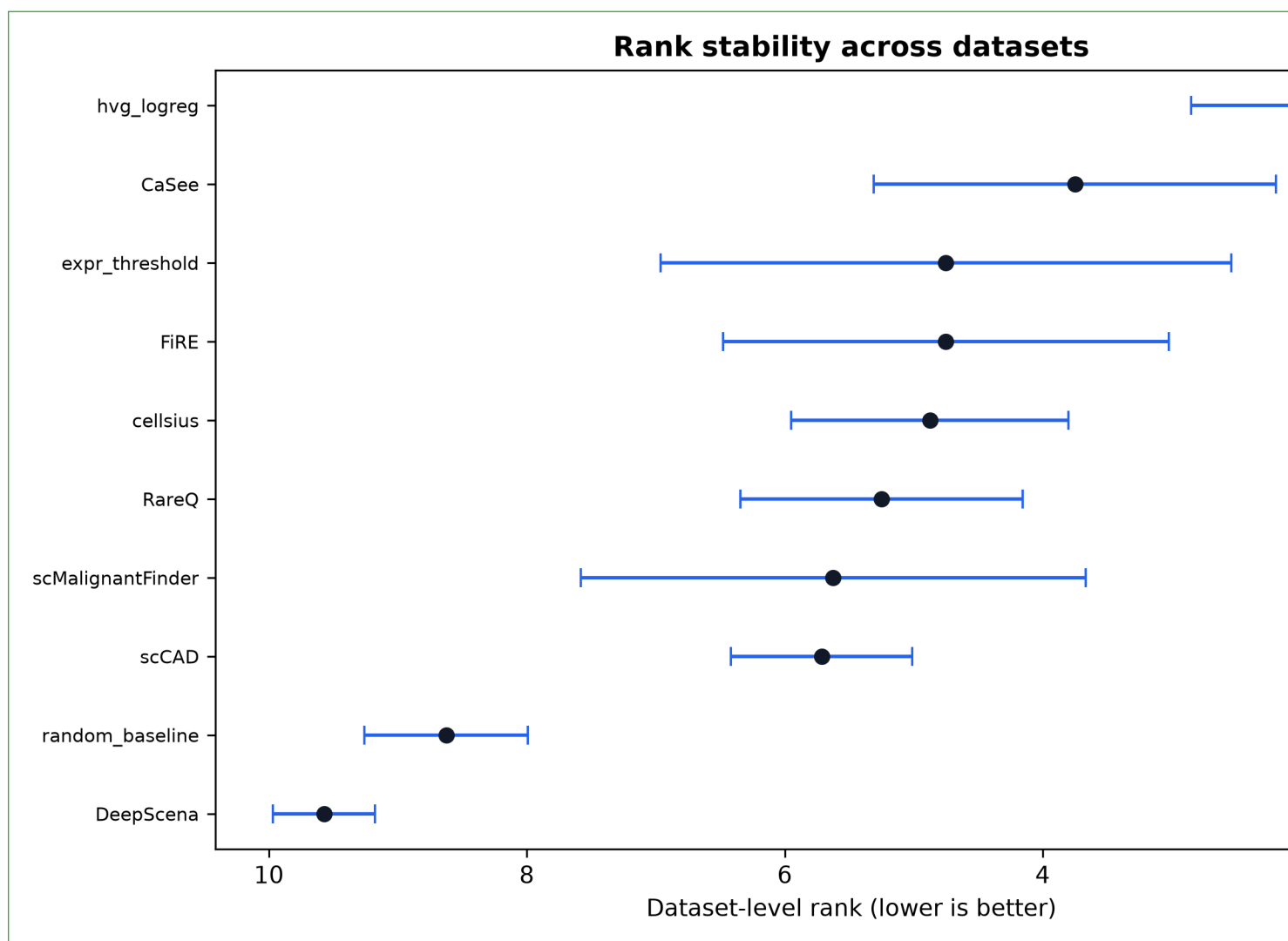
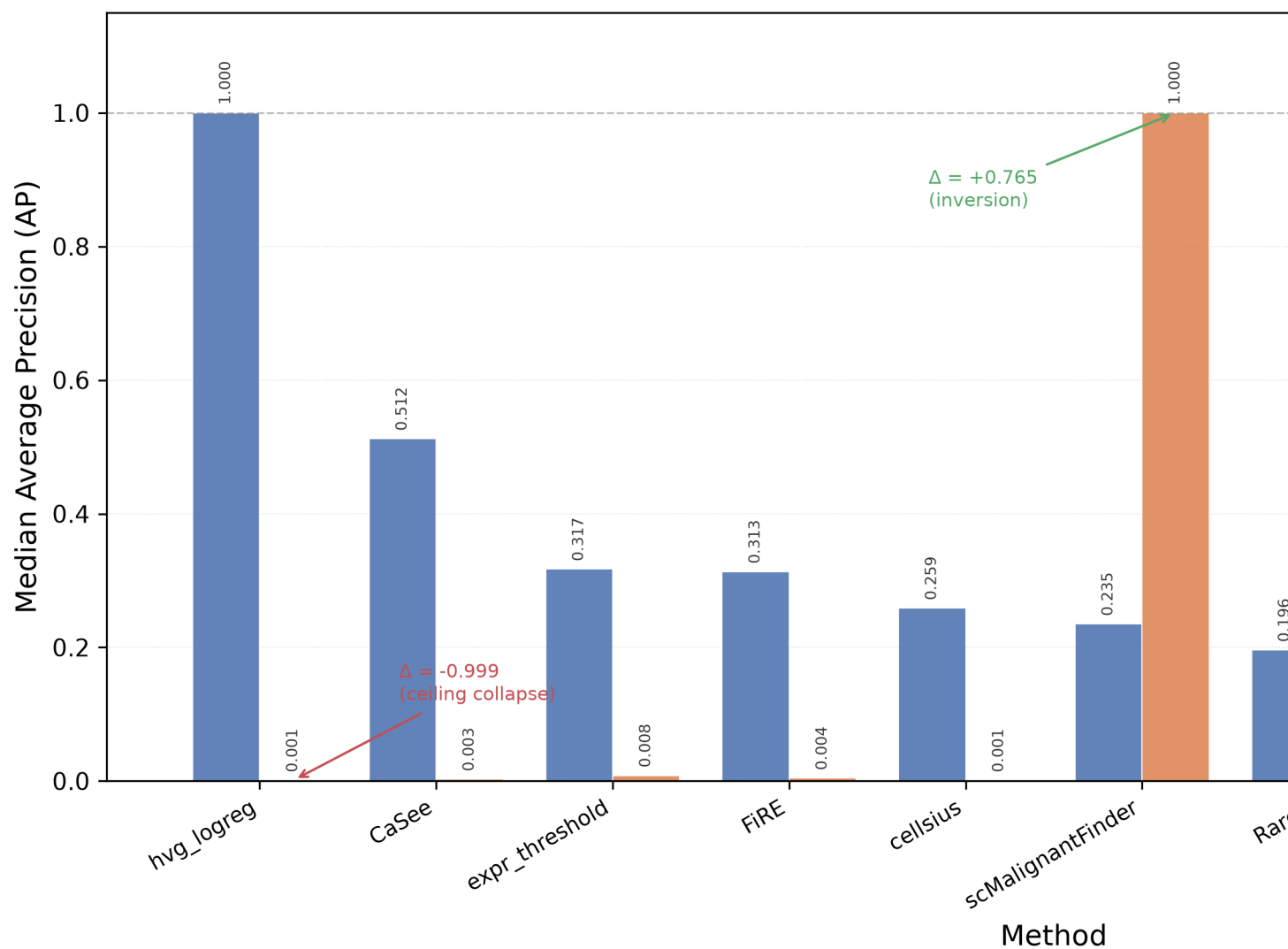


Figure 11: Figure 7. Rank bootstrap forest: rank stability across datasets.

Figure 11. Track F (Intra-lineage) vs Track A (Inter-lineage)
Malignant epithelial detection — crc_lee, N = 15 units (3 pre



Source: data/results/revision/track_f/track_a_vs_f_comparison.csv | All 10 methods evaluated (150/150 Track F units). Background = normal

Figure 12: Figure 11. Track F (intra-lineage) vs Track A (inter-lineage) median AP per method. hvg_logreg ceiling collapses from 1.000 to 0.001; scMalignantFinder inverts (0.235 to 1.000).

HC label set. The rank ordering under CNV-free labels correlates with the published ranking (Spearman $\rho = 0.648$, $p = 0.043$; Iman-Davenport $F = 171.01$, $p < 0.001$ [?, ?]). ~~The results of the pairwise Wilcoxon tests relative to FiRE (Table 6) suggest that FiRE has significantly better performance than random baseline~~data/results/revision/circularity_ranking.csv), but this correlation is computed over near-tied chance-level values, so it should be interpreted cautiously: the rank order is broadly preserved, but the absolute performance differences vanish without CNV. We therefore frame CNV as the dominant but not sole label component — tissue-of-origin and signature scores contribute to the HC set but CNV drives the discriminability. We also validated inferred CNV burden (mean absolute deviation from the diploid baseline) against published source malignancy annotations across 9 datasets (57,811 cells; 5,DeepSeena, seCAD010 positive; 52,RareQ and cellsius ($p < 0.05$ after Benjamini-Hochberg FDR correction)). ~~There are no significant differences between seMalignantFinder and exprthreshold methods when compared to FiRE. In our exploratory comparison, CaSee is better than FiRE (Δ mean AP 801 background; data/results/revision/cnv_concordance.csv).~~ CNV burden discriminated source-annotated malignant from non-malignant cells with overall AUROC = ~~-0.1400~~0.782 and MCC = 0.361. Per-dataset AUROC ranged from 0.747 (hcc_wei) to 0.960 (luad_laughney), with five datasets exceeding 0.86 (luad_laughney 0.960, pdac_peng 0.934, crc_lee 0.892, hnscc_puram 0.906, ov_izar_tirosh 0.868). An earlier version of this analysis reported hnscc_puram at AUROC = 0.500 because its cell_type labels (e.g. "0.0", ~~BH FDR~~"Fibroblast", "T cell") did not match infercnvpy's default reference categories (T cells, B cells, Myeloids, ...) and CNV scores were zero; we corrected the reference categories to include the singular label forms present in this dataset and regenerated its CNV, yielding AUROC = 6.1×10^{-8} ; Cliff's $\delta = -0.245$) ~~; this is reported under the exploratory versus published category, not under the primary ranked method category [?]~~. 0.906 (MCC = 0.763). CNV is therefore used as one of three independent HC-label components, not as a standalone ground truth. Cross-validation with CopyKAT, Numbat, or SCEVAN would further strengthen CNV evidence but was not feasible at this scale.

2. **Imperfect ground truth.** High-confidence labels (P_HC/B_HC) rely on consensus across four evidence arms. False positives in source annotation or CNV calls can propagate. The tier-assignment system mitigates this by using only P_HC vs B_HC for primary AP computation.
3. **Fallback and degenerate rates.** Across all methods, 249 units produced fallback scores and 377 units produced degenerate outputs. These rates are high and reflect current engineering stability limitations of method wrappers rather than algorithmic quality.
4. **Dataset diversity.** The 10 datasets span 8 solid-tumour types and 2 blood malignancies but are dominated by 10x Chromium (8/10). SMART-seq2 is represented in only 2 datasets. No spatial transcriptomics or multi-omics data are included.
5. **Track D size.** Natural prevalence evaluation (Track D) contains only 30 units from 2 datasets, limiting statistical power for this track.
6. **No held-out datasets.** All datasets are public. Method developers can tune against the leaderboard, though REACH's multi-track design with null controls and label-noise tracks makes simple overfitting harder.
7. **Single contributor.** The benchmark was developed by a single author. Independent verification by additional researchers would strengthen reproducibility claims.
8. **Cancer-type scope.** Several therapeutically important cancer types (glioblastoma, prostate adenocarcinoma, gastric cancer) are not represented.
9. **Label-threshold sensitivity.** To test whether the qualitative method ordering depends on the AUCell score threshold and kNN neighbourhood-purity cutoff, we varied both across nine combinations (AUCell 0.10/0.15/0.20 $\times$ kNN 0.40/0.50/0.60) while holding CNV scores fixed, and recomputed rankings. On the five datasets yielding a valid Spearman ρ , the high-confidence label sets were insensitive to these threshold variations — the same cells were selected at every setting — so the resulting method rankings are identical within each dataset (per dataset: luad_laughney 0.685, pdac_peng 0.758, bcc_yost 0.818, hcc_wei 0.879, crc_lee 0.939). The cross-dataset range [0.685, 0.939] therefore reflects dataset-level variation in label separability, not threshold sensitivity. Two datasets (ov_izar_tirosh, rcc_multi) could not form a high-confidence background at any setting ($n_{B_HC} = 0$) and are excluded. The label-construction pipeline is thus stable to the tested threshold parameters, though this analysis does not demonstrate ranking robustness to threshold changes that would actually alter the label set (data/results/revision/threshold_sensitivity.csv).
10. **TME overlap and the false-positive scope choice.** Because REACH measures detection against a heterogeneous TME background, methods trained or implicitly calibrated on the malignant-vs-TME con-

trast may flag TME subpopulations whose expression partially overlaps malignant signatures (e.g. activated macrophages, plasmablasts, cycling fibroblasts) as false positives. This is a deliberate scope choice: REACH models the realistic low-prevalence screening scenario in which TME is the background and tolerance to such false positives is modulated by precision@k. Track C confirms that, on pure all-negative null units, every unsupervised method keeps its mean FPR near 3% (`data/results/revision/sens2_sens3_extract.txt`), so elevated FPR on mixed-TME units reflects biological overlap, not metric pathology. The new intra-lineage Track F (R3-4/R3-5) begins to disentangle behaviour on pure-epithelial versus mixed-TME backgrounds.

~~Selected pairwise comparisons on Track A AP from `pairwise_tests.csv`. Positive Δ means FiRE has higher mean AP than the comparator; negative Δ means the comparator has higher mean AP.~~

1.5 Comparison to Related Benchmarks

~~$\llbracket @ > p() * 0.1304 > p() * 0.1739 > p() * 0.3043 > p() * 0.3913 @$~~

~~Comparator Δ -mean AP- Feature~~

~~Wilcoxon p -REACH v1.2~~

~~BH-FDR-scIB (Luecken 2022)~~

~~Cliff's δ -OpenProblems (Lance 2024)~~

~~DeepScena 0.3320-~~

~~Task 1.10×10^{-24} -Rare malignant cell detection 1.66×10^{-23} -Batch correction, clustering, integration 0.7539 Multi-task (denoising, DE, etc.)~~

~~random_baseline 0.3226-Tracks 2.52×10^{-22} 6 (spike-in, synthetic, null, natural, noise, intra-lineage) 8.11×10^{-22}~~

~~Multi-metric per task 0.7271-Per-task metrics~~

~~GaSee -0.1395-Labels 3.96×10^{-8} -Multi-arm confidence tiers (P_HC-B_LC) 6.14×10^{-8} -Canonical cell-type labels -0.2451-Truth from data generators~~

~~scCAD 0.1350-Null controls 1.56×10^{-4} -Yes (Track C) 2.34×10^{-4} -Not standard 0.2886-Not standard~~

~~RareQ 0.1161-Label noise 1.77×10^{-3} -Yes (Track E, supervised only) 2.49×10^{-3} -Not included 0.2457-Not included~~

~~cellsius 0.0978-Supervised ceiling 3.06×10^{-3} -Yes (hvg_logreg) 4.04×10^{-3} -Not standard 0.1930-Not standard~~

~~scMalignantFinder0.0453-Fallback handling 0.812-Filtered from primary AP 0.831-Not standardised 0.1206 Varies~~

~~expr_threshold 0.0304-Containerisation 0.836-Docker + GHCR 0.836-Docker 0.0816-Docker + Nextflow~~

The bootstrap intervals for ranks are stable, whereas the hvglogreg and CaSee methods are ranked 1 and 2 respectively for every resample of 2,000. On the other hand, FiRE is observed to have a median rank of 3 (with a 95% bootstrap rank interval of 3, 4). Expr threshold, scMalignantFinder, cellsius, RareQ, and scCAD fall into an overlapping middle band (Fig. ??(a)). The mean within-unit rank for each method, shown in Fig. ??(b), separates the strongest methods from the overlapping middle group and is consistent with the bootstrap rank intervals [?].

1.6 Rarity sensitivity and track specific robustness

Rarity sensitivity was assessed by regressing nAP against $\log_{10}(\pi_u)$, where π_u denotes the positive-cell prevalence in each benchmark unit (Fig. ??(c)). Most methods show positive slopes, indicating that performance improved as rare cells became less sparse. CellSIUS shows the largest nAP slope (0.220

1.6 Data and Code Availability

- **Source code:** <https://github.com/jaswanthmoram/reach-rarecell-benchmark>
- **Concept DOI:** <https://doi.org/10.5281/zenodo.19847108>
- **Processed datasets (7.3 GB):** <https://doi.org/10.5281/zenodo.19850652>
- **Track Units A-C (9.7 GB):** <https://doi.org/10.5281/zenodo.19850972>
- **Track Units D-E (2.2 GB):** <https://doi.org/10.5281/zenodo.19851287>
- **Complete results (425 MB):** <https://doi.org/10.5281/zenodo.19851710>

- **Docker image:** ghcr.io/jaswanthmoram/reach-rarecell-benchmark:latest
- **License:** MIT
- **GEO accessions:** GSE103322, $p = 1.78 \times 10^{-20}$ GSE123813, while RareQ, CaSee, seMalignantFinder, seCAD, FiRE, and exprthreshold show similar positive slopes. These nAP slopes are descriptive [?, ?]. There is a reordering of methods in Track B versus Track A (Spearman rank correlation $\rho = 0.042$ GSE149614, $\rho = 0.907$, so the Track B stands as a stress test versus being incorporated into the primary leaderboard (Fig. ??(a), left side) [?]. The track E hvg-logreg label noise analysis shows that the loss in mean AP is relative to a Track A mean AP of 0.9088: hvg-logreg mean AP drops between 0.909 and 0.534 under 30% background-to-positive flips, 0.829 under 30% positive to background flips, 0.481 under 10% symmetric noise and 0.460 under 20% symmetric noise. On the other hand, the corresponding robustness indices (ranging from 0 to 1) for these analyses are 0.588 GSE123902, 0.912 GSE202051, 0.529 GSE132465, 0.507 respectively (Fig. ??(a), right side) [?]. GSE159115, GSE146026, GSE161801, GSE109761

1.7 Null behavior and calibration

Git alone is sufficient for toy workflows and public snapshot reproduction. Full processed-data, track-unit, prediction, and complete-result reruns require restoring the external Zenodo archives listed above.

Track C evaluates false-positive behavior in background dominated units, where no true rare malignant cells are expected (Fig. ??(b)) [?]. For each method, REACH measures the fraction of top ranked cells that are incorrectly selected as positives under these null-control conditions. Most unsupervised, exploratory, and naive scoring methods show an average top k false positive rate of approximately 3% : CaSee = 0.0301, DeepScena = 0.0306, FiRE = 0.0315, RareQ = 0.0293, CellSIUS = 0.0295, expr-threshold = 0.0310, seCAD = 0.0301, seMalignantFinder = 0.0315, and random baseline = 0.0309. In contrast, the supervised hvg-logreg ceiling shows a much higher Track C false positive rate of 0.788. This behavior is expected because hvg-logreg is an in-sample supervised ceiling, not a deployable unsupervised detector. Calibration has been assessed only for methods that produced probability-like scores [?, ?].

1.7 Runtime, scalability, and failure modes

1.7 Reproducibility

Runtime varies by approximately two orders of magnitude across methods (Fig. ??). Track A Mean Execution Times (MET) for the corresponding tools range from 3.3 seconds (hvg-logistic regression) to 7.6 seconds (seMalignantFinder) to 357.5 seconds (CaSee) ; FiRE MET of 118.1 seconds, seCAD MET of 123.4 seconds, RareQ MET of 78.2 seconds, DeepScena MET of 80.6 seconds, cellsius MET of 39.8 seconds [?, ?]. Runtime also needs to be put into context with fallback and/or degenerate output behavior. Runtime should therefore be interpreted together with fallback and degenerate output rates, because a fast method is not necessarily reliable if it frequently returns constant scores or fails on many benchmark units [?, ?].

1.7.1 Snapshot Reproduction (Git-only, no external data)

Rank uncertainty, critical difference, and rarity sensitivity. (a) Bootstrap rank confidence intervals. Ranks are computed from Track A AP over 2,000 unit level bootstrap resamples (rank_ci.csv). Lower rank is better. (b) Critical difference rank view. It shows the mean within unit ranks on the common Track A subset, with the Nemenyi critical difference bar [] rcb smoke-test python scripts/run_all.py --toy python scripts/phase11_statistics.py --from-snapshots python scripts/reproduce_from_snapshots.py snakemake -n --cores 1 dvc repro --dry

See docs/reproducibility_receipt.md for the verified execution record, environment details, and expected outputs.

1.7.2 Revision compute environment

The sensitivity, CNV-validation, Track F, and concordance analyses in this revision were executed on a single CPU-only node: AMD EPYC 7B12 (CD ≈ 1.14 , $k = 10$, $N = 14016$ vCPUs), 62 GiB RAM, Debian GNU/Linux 13 trixie, Python 3.13.5, R 4.5.0, with scanpy 1.12.1, $\alpha = 0.05$ scikit-learn 1.9.0, infercnvpy 0.6.1, anndata 0.11.4, torch 2.12.1 (CPU). R packages FiRE, CellSIUS, RareQ, and Seurat were installed and ran natively. **No GPU was available on this node.** DeepScena, whose TensorFlow/Torch pipeline requires CUDA, was run separately on a Google Colab Tesla T4 GPU; its predictions were transferred back and scored identically to all other methods. Full environment capture: data/results/revision/environment_capture.txt. (The original v1.2.0

preprocessing cited in the Acknowledgements used GCP high-memory CPU and NVIDIA L4 GPU instances; that remains the record for dataset pre-processing, while the revision compute above is the record for the new analyses.) **(c)** Rarity sensitivity. It depicts AP and normalized AP across Track A prevalence tiers T1-T4. Several methods degrade sharply in ultra-rare units, while the supervised ceiling remains near 1 across tiers.

Track sensitivity, robustness, and null behavior. **(a)** Track sensitivity and supervised label noise diagnostics. Left: Track A vs Track B mean AP per method, with the diagonal showing equal performance. Right: Track E robustness indices for the supervised ceiling under four label perturbations. Track E is not interpretable for unsupervised methods. **(b)** Track C null behavior and calibration. Background-dominated units assess false positive behavior. ECE is shown only for methods with probability-like outputs.

1.8 Author

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2 Discussion

REACH shows a clear gap between supervised separability and the performance of current unsupervised rare malignant-cell ranking methods [?, ?]. The supervised hvg-logreg ceiling indicates that many high confidence malignant and background labels contain recoverable gene expression differences. In other words, when label information is available, these cell groups can often be separated with very high accuracy. However, the lower Track A AP values obtained by unsupervised and rarity-based methods show that current scoring approaches recover only part of this separability [?, ?]. This gap is important because rare malignant cell detection is not only a question of whether the signal exists in the data, but also whether an unsupervised method can reliably recover it under strong class imbalance and tumor heterogeneity.

1.1 Author Contributions

The results also support the need for a multi-track benchmark rather than a single leaderboard. Track A provides the primary evaluation under controlled real-cell rarity. Track B tests whether methods remain stable under synthetic stress, but its ranking does not reproduce Track A and therefore should not be averaged into the primary leaderboard. Track C adds a null control setting to examine false-positive behavior when malignant cells are not expected. Track D preserves natural prevalence in blood and circulating tumor cell datasets. Track E evaluates the effect of label corruption on M.V.S.J. conceived the benchmark, designed the supervised ceiling and should not be interpreted as evidence of unsupervised robustness [?, ?, ?]. Together, these tracks show that different evaluation conditions reveal different aspects of method behavior.

The main practical message from REACH is that method performance should be interpreted with caution. FiRE is the strongest ranked published comparator on the primary Track A endpoint, but its advantage is not uniform across all comparisons. In the paired tests, FiRE is not significantly better than the naive expr-threshold baseline or seMalignantFinder. CaSee performs better than FiRE in the exploratory comparison, but it is substantially slower and produces constant scores in some benchmark units [?, ?]. Dataset level results also show that different methods perform best in different tumor contexts. Therefore, REACH does not support the idea of a single universally best method. Instead, rare malignant cell detection methods should be reported together with the dataset context, prevalence level, failure behavior, and the evaluation track in which they were tested [?, ?, ?].

Runtime and AP Pareto behavior. Left: per unit runtime versus cells per unit. Right: mean Track A runtime versus mean Track A AP, highlighting the trade-off between ranking performance, compute cost, and method stability. evaluation framework, curated datasets, implemented the software, executed the analyses, generated figures, interpreted results, and wrote the manuscript.

2 Limitations

This study has a few limitations that should be considered when interpreting the results. First, CNV evidence in the current version of REACH was generated using infercnvpy as the primary CNV inference approach. CopyKAT was not used as a primary scored comparator because it did not satisfy the uniform cell-level score contract and could not be applied consistently across all datasets. This means that CNV evidence was handled consistently within the benchmark, but it was not independently cross-validated using a second CNV caller [?, ?].

1.1 Acknowledgements

Second, the current dataset collection does not yet cover the full diversity of cancer scRNA-seq data. Most datasets are based on 10x Chromium technology, while only two datasets are based on Smart-seq2. Spatial transcriptomics, multi-omics datasets, The author thanks the original dataset creators and method developers whose public data and software made REACH possible. This work used Google Cloud Platform high-memory CPU instances (30 vCPU / 240 GB RAM and 60 vCPU / 480 GB RAM) and NVIDIA L4 GPU instances for preprocessing, track generation, method execution, and a wider range of tumor types are not included in this version [?]. Therefore, the present results should be viewed as a first benchmark snapshot rather than a complete evaluation of all possible rare malignant cell detection settings. result aggregation. The author also acknowledges the open-source Scanpy, scikit-learn, Snakemake, DVC, Docker, and Python scientific-computing communities.

Third, some degree of circularity may remain in the label construction. Because CNV evidence contributes to the definition of high-confidence malignant cells, methods that indirectly capture CNV-related expression patterns may benefit in some datasets. This is particularly relevant for datasets such as *ov_izar_tirosh*, where CNV-supported labelling contributes to the evidence used for defining malignant cells. Future versions of REACH should reduce this limitation by comparing multiple CNV callers and by testing label definitions that are less dependent on a single evidence source. Degenerate and failed method runs (CopyKAT, MACE, SCANNER, SCEVAN, RaceID3, scATOMIC, GiniClust3) were identified on these systems and documented in the excluded-methods table.

Fourth, Track E has a limited interpretation. It is useful for examining how the supervised hvg-logreg ceiling responds to corrupted labels, but it should not be interpreted as a general robustness test for unsupervised rare cell detection methods. This distinction is important because unsupervised methods do not use label information in the same way as the supervised ceiling.

Fifth, method stability remains an important practical issue. In the current benchmark, 249 fallback units and 377 degenerate units were recorded. These failures show that some methods still face difficulties related to memory use, runtime limits, wrapper stability, or constant score outputs. For rare cell detection benchmarks, reliability is therefore as important as accuracy: a method that performs well in some units but fails in many others may not be practically useful [?, ?].

1.2 Funding

Finally, public leaderboards can create a risk of overfitting. If methods are repeatedly tuned on the same visible datasets and tracks, leaderboard performance may not reflect general performance on unseen data. Future versions of REACH should include hidden test datasets, pre-registered track definitions, post-submission perturbation tracks, and containerized execution with fixed prediction schemas. The aforementioned changes could make the benchmark more robust and more useful for evaluating new rare malignant cell detection methods [?, ?]. No external funding was received for this work.

2 Conclusion

REACH offers a reproducible way to compare methods for detecting rare malignant cells in scRNA-seq data. The benchmark is designed around the practical difficulties of this task: rare cells are often present at very low frequency, their labels may require more than one source of evidence, and method outputs can vary across datasets and tumor contexts. To address these issues, REACH combines multi-evidence high-confidence labels, controlled and natural prevalence settings, null controls, label noise analysis, fallback-aware metrics, statistical testing, and reproducible figure generation from audited results.

1.1 Competing Interests

The author declares no competing interests.

The current benchmark shows that rare malignant cell detection remains challenging. The supervised hvg-logreg ceiling performs strongly on Track A, suggesting that many high confidence malignant and background cells can be separated using gene expression information. However, current unsupervised and rarity-based methods recover only part of this separation. Among the ranked published methods, FiRE gives the strongest performance on the primary Track A endpoint, while CaSee provides a strong exploratory comparison but should not be treated as an ordinary ranked published detector. The results also indicate that the best performing method changes from one dataset to another, suggesting that no single method is consistently best across all tumor contexts.

These findings support the need for careful, context specific benchmarking. Instead of reducing performance to one overall leaderboard, REACH encourages results to be interpreted together with tumor type, rarity level, false-positive

1.2 Supplementary Material

- Excluded-method notes: `data/results/manuscript/supplementary/EXCLUDED_METHODS.md`
- Sensitivity-analysis notes: `data/results/manuscript/supplementary/SENSITIVITY_ANALYSES.md`
- Statistical-method notes: `data/results/manuscript/supplementary/STATISTICAL_METHODS.md`
- Phase 11 tables: `data/results/tables/phase11/`
- Phase 12 figures: `data/results/figures/phase12/`

1.3 Contribution to the Field

Rare malignant cells are central to tumor progression, treatment resistance, and circulating tumor-cell biology, but computational methods for detecting them in scRNA-seq data are difficult to compare fairly. REACH addresses this gap by providing a reproducible, cancer-focused benchmark that evaluates rare malignant-cell detection as a ranked retrieval problem rather than a standard balanced classification task. The benchmark combines multi-evidence confidence labels, controlled real-cell rarity, synthetic stress tests, null-control behavior, ~~method stability, and evaluation track.~~ In this sense natural-prevalence evaluation, label-noise analysis, standardized method wrappers, and explicit failure auditing. By releasing code, figures, tables, Docker assets, and Zenodo archives, REACH provides a transparent ~~platform for future method comparison~~ baseline for method developers and a practical ~~reference for developing more robust and reproducible rare malignant cell detection methods~~ guide for researchers selecting rare malignant-cell detection tools.

Data and Code Availability

The REACH benchmark framework, including all source code, pipelines, and evaluation scripts, is publicly available at: [https://github.com/REACH-benchmark](#). Versioned Zenodo archives containing the processed datasets, benchmark units, and result tables are available with the latest release. All datasets used in this project have been obtained from public databases, specifically the GEO, and all sources are cited within the manuscript. The processed dataset, along with benchmark units and performance results, will also be available with the repository. The GitHub repository includes toy workflows, scripts for deterministic snapshot regeneration, and documentation for reproducing the benchmark using the archived data and specified compute environment. All results/figures reported in this analysis can be regenerated deterministically from any prediction output stored using the scripts provided. All environment specifications are thoroughly documented in order to achieve reproducibility across computational environments.

1.4 Version History

Acknowledgements

- **v1.2.0 (2026-04-29):** Publication-ready snapshot. 10 datasets, 10 methods, 1,125 units, 11,250 evaluations. Added public Phase 11/12 tables and figures, label-based evaluation, all method wrappers exposed through registry, paper.md, and reproducibility receipt.

- **v1.1.0 (2026-04-28):** Initial public release. Code, configs, tests, Docker, CI/CD, toy-data generation, frozen CSV snapshots, Zenodo DOIs.